

not a functional protein make. All proteins depend on their three-dimensional structure to carry out their function. Molecules called *chaperonins* aid in the folding process (either in the cytoplasm, ER, or golgi). Once a protein is assembled into its final shape (if being prepared for export), it is sorted and packaged into vesicles by the time it reaches the end of the golgi apparatus.



TIP

Don't let the labels confuse you. Proteins are chains of amino acids (usually very long chains of at least 100 acids). Enzymes, used to catalyze reactions, also are chains of amino acids and therefore also are categorized as proteins. *Polypeptides*, or simply *peptides*, are shorter chains of amino acids used to bond larger protein molecules, but they also can be regarded as proteins. Both antibodies and many hormones are also proteins, along with almost everything else in the body — hair, muscle, cartilage, and so on. Even the four basic blood types — A, B, AB, and O — are differentiated by the proteins (antigens) found in each.

45-54 Fill in the blanks to complete the following sentences:

Protein synthesis begins in the cell's 45._____ when the gene for a certain protein is 46._____ into messenger RNA, or mRNA, which then moves on to the 47. _____. There, ribosomes 48._____ three base pairs at a time, forming a series also referred to as a 49._____. Molecules of 50._____ then collect each amino acid needed so that the ribosomes can link them together through 51._____ bonds. The chain is then folded into its 3D structure with the help of 52._____. Now in the 53._____ the protein is packaged into a 54._____ for transport to the cell membrane to be released.

55 Which of the following comes first in the protein-synthesis process?

- a. Transfer RNA
- b. Transcription
- c. Peptide bonds
- d. Messenger RNA

e. Translation

56 What is a codon?

- a. A sequence of three adjacent nucleotides on a mRNA molecule
- b. A type of amino acid used in the cellular production of protein
- c. A specialized ribosome required to translate DNA
- d. A base pair that connects complementary strands of mRNA and tRNA

Answers to Questions on the Cell

The following are answers to the practice questions presented in this chapter.

- ① Cell that can become any type of other cell: **e. totipotent**
- ② Partially differentiated stem cell; can still become numerous types but is limited: **c. multipotent**
- ③ Sequence of events that leads to the death of a cell: **a. apoptosis**
- ④ Cell that carries out a specific specialized function: **b. differentiated**
- ⑤ Cell that can become most other cells, except placenta: **d. pluripotent**
- ⑥-⑨ The lipid bilayer structure of the cell membrane is made possible because phospholipid molecules contain two distinct regions: The **6. hydrophilic** region is attracted to water. The **7. heads** of the phospholipids thus create an internal and external membrane surface. The **8. hydrophobic** region is repelled by water, so the **9. tails** of the phospholipids create an internal layer that will not interact with water.
- ⑩ What is a hypotonic solution? **b. A solution that has a greater concentration of water than exists in the cell.**



TIP

The prefix *hypo* refers to under or below normal. The prefix *hyper* refers to excess, or above normal. Someone who has been out in the cold

too long suffers hypothermia — literally insufficient heat. So a solution, or tonic, with very few particles would be hypotonic.

- ⑪ Injecting a large quantity of distilled water into a human's veins would cause many red blood cells to **a. swell and burst**. With a hypotonic solution outside the cell, the membrane would allow osmosis to continue past the breaking point. **Remember:** Hypo makes cells big, like a hippo.
- ⑫ Which of the following processes requires energy from ATP? **c. II only**. Endo- and exocytosis are the only ones on the list that require energy. While facilitated diffusion does require a protein for molecules to travel through, it does not need energy.
- ⑬ Droplets of liquid are brought into the cell: **c. pinocytosis**
- ⑭ Cell membrane extends around a molecule engulfing it into the cell: **d. phagocytosis**
- ⑮ Molecules are pumped from areas of low concentration to high concentration: **a. active transport**
- ⑯ Molecules are released from the cell by a vesicle fusing with the cell membrane: **b. exocytosis**
- ⑰ First, receptor mediated endocytosis occurs to bring the molecule into the cell. Since the cell membrane spreads around the molecule, it enters already packaged in a vesicle. Rather than unpack its contents, the cell simply pushes the vesicle to the opposite end where it fuses with the cell membrane, releasing the molecule via exocytosis.
- ⑱ What is the role of the nuclear lamina? **e. to break down and rebuild the envelope**
- ⑲ In the nucleus of a cell that isn't dividing, DNA **b. is packaged within chromatin threads**.
- ⑳ The nucleolus **d. assembles ribosomes**. It's not just a coincidence that the nucleolus sits at the heart of genetic storage.
- ㉑ What is the function of a mitochondrion? **b. To produce energy through**

aerobic respiration. Each of the other options describes a function served by a different organelle.

- 22) Actin, which is the most abundant protein in human cells, **a. forms filaments that contribute to the cytoskeleton.** It makes sense that the protein making up much of the cytoskeleton is the most abundant because the cytoskeleton accounts for up to 50 percent of the cell's volume.
- 23) Which organelle translates genetic information to direct protein synthesis?
a. The ribosome



TIP

When you think of a big protein-laden meal, you think of ribs. Ribs. Ribosome. Protein synthesis. Get it?

- 24) Which organelle is the site of protein synthesis? **d. Rough (granular) endoplasmic reticulum.** Ribosomes isn't a choice, and it's the attachment of them that makes the RER rough.
- 25) Which of the following can change the consistency of cytoplasm? **e. All of the above.** In addition, agitation, or just being jostled, can change cytoplasmic consistency.
- 26-38) Following is how [Figure 3-3](#), the cutaway view of the cell and its organelles, should be labeled.
26. **c. Cytoplasm**; 27. **m. Vesicle (formation)**; 28. **g. Nucleolus**; 29. **h. Nucleus**; 30. **k. Rough endoplasmic reticulum**; 31. **d. Golgi apparatus**; 32. **i. Cell (plasma) membrane**; 33. **b. Cilia**; 34. **f. Mitochondrion**; 35. **j. Ribosomes**; 36. **l. Smooth endoplasmic reticulum**; 37. **e. Lysosome**; 38. **a. Centriole**

- 39) Mitochondrion: **d. Site of cellular respiration**
- 40) Nucleolus: **e. Site of ribosome production**
- 41) Flagellum: **a. Long, whiplike organelle for locomotion**
- 42) Cytoplasm: **b. Fluidlike interior of the cell that may become a**

semisolid, or colloid

④③ Lysosomes: **c. Membranous sacs containing digestive enzymes**

④④ Vesicle: **f. Structure used for transporting molecules in, out, or around the cell**

④⑤-⑤④ Protein synthesis begins in the cell's **45. nucleus** when the gene for a certain protein is **46. transcribed** into messenger RNA, or mRNA, which then moves on to the **47. rough endoplasmic reticulum**. There, ribosomes **48. translate** three base pairs at a time, forming a series also referred to as a **49. codon**. Molecules of **50. tRNA** then collect each amino acid needed so that the ribosomes can link them together through **51. peptide** bonds. The chain is then folded into its 3D structure with the help of **52. chaperonins**. Now in the **53. Golgi apparatus** the protein is packaged into a **54. vesicle** for transport to the cell membrane to be released.

⑤⑤ Which of the following comes first in the protein synthesis process? **b. Transcription**. Remember that you have to transcribe before you can translate and the result of transcription is mRNA.

⑤⑥ What is a codon? **a. A sequence of three adjacent nucleotides on a mRNA molecule**. It takes three nucleotides to round up a single amino acid.

Chapter 4

Divide and Conquer: Cellular Mitosis

IN THIS CHAPTER

- » Following the steps of cell division
 - » Understanding the results of errors in mitosis
-

Ever had so many places to be that you wished you could just divide yourself in two? If only we could be more like the cells that make us up! Cell division is how one “mother” cell becomes two identical twin “daughter” cells. Cell division takes place for several reasons:

- » **Growth:** Multicellular organisms, humans included, each start out as a single cell — the fertilized egg. That one cell divides (and divides and divides), eventually becoming an entire complex being.
- » **Injury repair:** Uninjured cells in the areas surrounding damaged tissue divide to replace those that have been destroyed.
- » **Replacement:** Cells eventually wear out and cease to function. Their younger, more functional neighbors divide to take up the slack.
- » **Asexual reproduction:** No, human cells don’t do this. Only single-celled organisms do.

Cell division occurs over the course of two processes: *mitosis*, which is when the chromosomes within the cell’s nucleus duplicate to form two daughter nuclei; and *cytokinesis*, which takes place when the cell’s cytoplasm divides to surround the two newly formed nuclei. Although cell division breaks down into several stages, there are no pauses from one step to another. Cell division as a whole is often referred to as mitosis because most of the changes occur during that process. Cytokinesis doesn’t start until later, but mitosis and cytokinesis do end together.

Keep in mind: Cells are living things, so they mature, reproduce, and die. They are not always actively engaged in the reproduction part. In this chapter, we review the entire cell cycle, and you get plenty of practice figuring out what happens when and why.

Walking through the Cell Cycle

The time that precedes cell division is referred to as *interphase*, and the majority of a cell's life is spent here. The length of time is highly variable as our cells have drastically different programmed life spans. Certain types of white blood cells and cells that line parts of our digestive tract often don't live to see 24 hours. Others, such as neurons and cardiac muscle cells, have the same life span as you. Regardless, each cell starts its life at the beginning of interphase and must go through two checkpoints before beginning the division process. After living part of their useful lives carrying out their specialized function, they reach the *restriction checkpoint*. From here, there are three options — though the choice is genetically programmed:

- » **Undergo apoptosis:** *Apoptosis*, or programmed cell death, is the fate of cells that have reached their life span (like an expiration date). Enzymes trigger this process and the cell is dismantled from the inside out (though it looks more like they're exploding).
- » **Remain specialized:** Cells continue to carry out their functions but do not divide. There's usually no going back from here; the cells live but never divide. This is the case for neurons, most muscle cells, and most white blood cells, among others.
- » **Prepare for division:** Interphase does not end here. There's still work to be done before cell division can begin.

Waiting for action: Interphase

Interphase is the period when the cell isn't dividing. It begins when the new cells are done forming and ends when the cell is prepared to divide. Although it's also called a "resting stage," there's constant activity in the cell during interphase, which is broken down into three subphases:

- » **G₁:** Stands for "growth" or "gap." During G₁, the cell grows and

develops, begins metabolism, and synthesizes proteins. The restriction checkpoint marks the end of this phase.

- » **S:** Stands for “synthesis.” *DNA synthesis* or *replication* occurs during this subphase. Each double-helix DNA molecule inside the cell’s nucleus is copied, creating two *sister chromatids*. At the end of S phase, the cell has two copies of every piece of DNA (chromosome) so that each daughter cell can receive a full set.
- » **G₂:** Stands for “gap.” Enzymes and proteins needed for cell division are produced during this subphase. The cell continues to grow, more rapidly this time, and often additional organelles are made to dole out to the daughter cells. The second checkpoint occurs here, halting the process if DNA damage is detected. Otherwise, the cell officially begins the division process, or *mitosis*.

Sorting out the parts: Prophase

As the first phase of mitosis, *prophase* is when structures of the cell’s nucleus begin to disappear, including the nuclear membrane (or envelope), nucleoplasm, and nucleoli. The two centrosomes (duplicated during S phase), each containing two centrioles, push apart to opposite ends of the cell, forming poles. The centrioles produce protein filaments that form *spindle fibers* between the poles as well as *asters* (or *astral rays*) that radiate from the poles into the cytoplasm.

At the same time, the *chromatin* threads (or pieces of DNA) shorten and coil, forming *chromonemata*. These wrap around proteins called *histones* to further condense the structure forming visible *chromatids*. The two identical chromatids are linked in the middle by a *centromere* creating the x-shaped *chromosome* you’re familiar with. Now the DNA is more manageable to move around and distribute equally between the two cells.

Dividing at the equator: Metaphase

Now that the nucleus is gone, the chromosomes migrate to the middle of the cell. They line up exactly along the *equatorial plane* with each centromere on the invisible center line. Lastly, the spindle fibers connect the centromeres to the centrioles on the opposite ends of the cell.

Packing up to move out: Anaphase

In anaphase, the centromeres are split apart by the centrioles reeling in the spindle fibers. The sister chromatids (identical copies of DNA) are pulled to the opposite poles. The cell begins to elongate and a slight furrow develops in the cytoplasm, showing where cytokinesis will eventually take place.

Pinching off: Telophase

Telophase occurs as the chromosomes reach the poles and the cell nears the end of division. The spindles and asters of early mitosis disappear, and each newly forming cell begins to synthesize its own nucleus to enclose the separated chromosomes. The coiled chromosomes unwind, becoming chromatin once again. There's a more pronounced pinching, or furrowing, of the cytoplasm into two separate bodies, but there continues to be only one cell.

Splitting up: Cytokinesis

Cytokinesis means it's time for the big breakup. The furrow, formed by a contractile ring that will divide the newly formed sister nuclei, migrates inward until it cleaves the single, altered cell into two new cells. Each new cell is smaller and contains less cytoplasm than the mother cell, but the daughter cells are genetically identical to each other and to the original mother cell, and will grow to normal size during early interphase. Telophase is still wrapping up at this point, thus both processes end at the same time.



TIP

Try remembering the phases like this:

Interphase = intermission, Prophase = prepare, Metaphase = middle,
Anaphase = apart, Telophase = twins, Cytokinesis = complete

Now try this warm-up question:



EXAMPLE

Q. True or false: Cells are dormant during interphase.

A. False. While it's also called a "resting stage" (a misnomer!), plenty is going on during interphase.

1 The G_1 subphase of interphase is

- a. the period of DNA synthesis.
- b. the most active phase.
- c. the phase between S and G₂.
- d. the phase immediately before mitosis begins.
- e. part of cell division.

2 What happens to DNA during interphase?

- a. It's duplicated during the S subphase.
- b. It's cut in two during the G₂ subphase.
- c. It unwinds rapidly during the G₂ subphase.
- d. It separates the opposite ends during the S subphase.
- e. It develops a protective film during the G₁ subphase.

3 During prophase, the nuclear membrane, or envelope,

- a. splits into thirds.
- b. develops a series of concave dimples.
- c. begins to disappear.
- d. changes color.
- e. strengthens its structure.

4 Which of the following is true for metaphase?

- a. The nuclear membrane reappears.
- b. The chromosomes move to the poles.
- c. The chromosomes align on the equatorial plane.
- d. It's composed of subphases G₁, S, and G₂.
- e. Chromatin coils up making it easier to move.

5 During metaphase, each chromosome consists of two duplicate chromatids.

- a. True
- b. False

6 Identify an event that DOES NOT happen during anaphase.

- a. Early cytokinesis occurs with slight furrowing.
- b. The cell goes through the growth subphase.
- c. Centrioles shorten the spindles fibers.
- d. Chromatids move to the outside of the cell.
- e. The centromeres split.

7 Which event DOES NOT occur during telophase?

- a. The chromosomes uncoil.
- b. The chromosomes reach the poles.
- c. The spindle fibers degenerate.
- d. The nuclear membrane reforms.
- e. The cell splits in two.

8 What structures disappear during telophase?

- a. Spindles and asters
- b. Nuclear membranes
- c. Nucleolei
- d. Cytoplasm
- e. Chromatin

9 Which is the correct order of phases of the cell cycle and mitosis?

- a. Prophase, metaphase, anaphase, telophase, interphase, cytokinesis
- b. Interphase, cytokinesis, telophase, prophase, anaphase, metaphase
- c. Cytokinesis, interphase, telophase, anaphase, metaphase, prophase
- d. Interphase, prophase, metaphase, anaphase, telophase, cytokinesis

e. Prophase, metaphase, cytokinesis, anaphase, telophase, metaphase

10-21 Use the terms that follow to identify the stages and cell structures shown in [Figure 4-1](#).

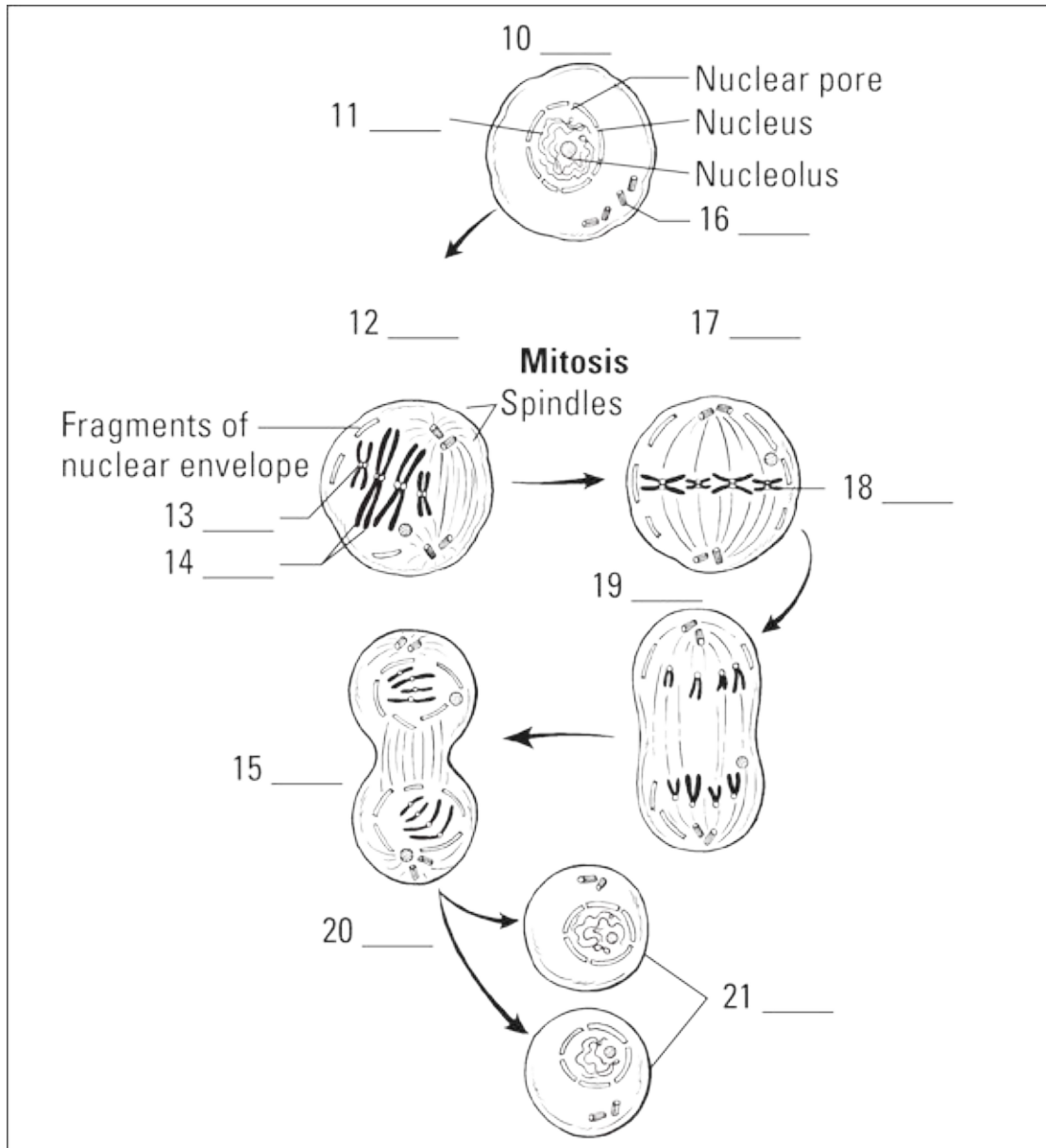


Illustration by Kathryn Born, MA

FIGURE 4-1: Cell structures and changes that make up the stages of the cell cycle.

- Anaphase
- Centromere

- c. Daughter cells
- d. Chromatin
- e. Cytokinesis
- f. Telophase
- g. Interphase
- h. Chromosomes aligned at equator
- i. Metaphase
- j. Centriole
- k. Prophase
- l. Sister chromatids

22 The two newly formed daughter cells are

- a. the same size as the mother cell.
- b. not genetically identical to each other.
- c. unequal in size.
- d. genetically identical to the mother cell.

23 Cytokinesis can be described as

- a. the period of preparation for cell division.
- b. the dividing of the cytoplasm to surround the two newly formed nuclei.
- c. the stage of alignment of the chromatids on the equatorial plane.
- d. the initiation of cell division.
- e. the dissolution of the cell's nucleus.

Understanding What Can Go Wrong during Cell Division

Mitosis is said to be a continuous process. That is, once it starts it doesn't stop; cells proceed through all the phases automatically. Though there are

checkpoints throughout (and even proofreaders to check DNA replication during S phase), some errors are inevitably missed. With the millions upon millions of cell divisions that happen in the human body, it's not surprising that sometimes things go wrong.

An error during DNA replication is referred to as a *gene mutation*. An error during mitosis leads to a *chromosome mutation*. As mentioned earlier, the DNA is checked for mismatched bases as it is being replicated. Mutations here can lead to a range of effects — from no discernable change, to a misfolded protein, to a completely non-functional gene. Chromosome mutations have a more drastic effect:

- » **Deletion:** Part of the chromosome just breaks off, taking the instructions of the gene with it.
- » **Insertion:** Additional copies of the genes are inserted into the chromosome.
- » **Inversion:** A section of the chromosome breaks off but is flipped upside down before reattaching.
- » **Nondisjunction:** An entire chromosome is pulled to the pole rather than splitting into sister chromatids. This leaves one daughter cell with three copies of each chromosome and the other with only one.

Often, in cases of chromosomal mutations especially, cells die during their first G1 phase because they are unable to perform all their necessary functions. Cells that survive can produce too much or too little of a protein affecting the relevant physiological process. If a mutation affects any of the numerous genes that control the cell cycle, mitosis can end up on fast-forward. Accelerated mitosis can lead to the formation of a tumor, also called a *neoplasm*. The rate of division usually restricts itself to replacing worn out or injured cells, but with accelerated mitosis, the cells don't know when to stop dividing.

- 24** What is the name of the mutation during mitosis that can result in an extra copy of chromosome 21, which causes Down syndrome?
- a. Neoplasm
 - b. Nondisjunction

- c. Insertion
- d. Inversion
- e. Acceleration

25 Accelerated mitosis can cause the growth of a tumor, which is also known as a(n)

- a. nondisjunction.
- b. daughter cell.
- c. neoplasm.
- d. kinetochore
- e. apoptosis.

26 An error that occurs during S phase is called a chromosome mutation.

- a. True
- b. False

Answers to Questions on Mitosis

The following are answers to the practice questions presented in this chapter.

- 1** The G1 subphase of interphase is **b. the most active phase**. With all that organelle-growing, metabolizing, and protein-synthesizing that takes place during G1, this isn't surprising.
- 2** What happens to DNA during interphase? **a. It's duplicated during the S subphase**. Remember, the cell is S-yntesizing new DNA molecules during this phase.
- 3** During prophase, the nuclear membrane, or envelope, **c. begins to disappear**. Otherwise, the spindle fibers wouldn't be able to reach the chromosomes.
- 4** Which of the following is true for metaphase? **c. The chromatids align on the equatorial plane**. Each of the other answer choices occurs during

earlier or later phases.

- 5) During metaphase, each chromosome consists of two duplicate chromatids. **a. True.** That way, each resulting daughter cell will have identical chromosomes.
- 6) Identify an event that DOES NOT happen during anaphase. **b. The cell goes through the growth subphase.** G_1 , the growth subphase, took place back in interphase.
- 7) Which event DOES NOT occur during telophase? **e. The cell splits in two.** This is a common mistake, especially since telophase and cytokinesis end at the same time. Remember that telophase is all about the nucleus, cytokinesis is all about the cell.
- 8) What structures disappear during telophase? **a. Spindles and asters.** These structures disappear because they're no longer needed at the end of mitosis.
- 9) Which is the correct order of phases of the cell cycle and mitosis? **d. Interphase, prophase, metaphase, anaphase, telophase, cytokinesis.**
- 10-21) Following is how [Figure 4-1](#), the stages and structures of mitosis, should be labeled.
10. **g. Interphase;** 11. **d. Chromatin;** 12. **k. Prophase;** 13. **b. Centromere;** 14. **i. Sister chromatids;** 15. **f. Telophase;** 16. **j. Centriole;** 17. **i. Metaphase;** 18. **h. Chromosomes aligned at equator;** 19. **a. Anaphase;** 20. **e. Cytokinesis;** 21. **c. Daughter cells**
- 22) The two newly formed daughter cells are **d. genetically identical to the mother cell.** None of the other answer options makes sense.
- 23) Cytokinesis can be described as **b. the dividing of the cytoplasm to surround the two newly formed nuclei.**
- 24) What is the name of the mutation during mitosis that can result in an extra copy of chromosome 21, which causes Down syndrome? **b. Nondisjunction.** Split the word into pieces. "Non" means negative, or it didn't happen. "Dis" means "apart" or "asunder." And you know that a "junction" is where things are joined. Down syndrome occurs when

chromosomes fail to split up properly, leaving daughter cells with the wrong number of chromosomes.

- 25 Accelerated mitosis can cause the growth of a tumor, which is also known as a(n) **c. neoplasm.**
- 26 An error that occurs during S phase is called a chromosome mutation. **b. False.** S phase is when DNA is being copied, so mutations here would affect the base pairing, a gene mutation. For a chromosomal mutation, the error has to occur during mitosis when the chromosomes are condensed.

Chapter 5

The Study of Tissues: Histology

IN THIS CHAPTER

- » Checking out epithelial tissue
 - » Keeping things together with connective tissues
 - » Flexing muscle tissues
 - » Sending signals through nervous tissue
-

Oh, what tangled webs we weave! As the chapter title says, *histology* is the study of tissues, but you may be surprised to find out that the Greek *histo* doesn't translate as "tissue" but instead as "web." It's a logical next step after reviewing the cell and cellular division (see [Chapters 3](#) and [4](#)) to take a look at what happens when groups of similar cells "web" together to form tissues. The four different types of tissue in the body are as follows:

- » Epithelial tissue (from the Greek *epi*– for "over" or "outer")
- » Connective tissue
- » Muscle tissue
- » Nervous tissue

In some tissues, like many of the connective tissues, the cells do not make physical contact with each other. The space between them can be filled with various fibers or simply left "empty," though that empty space would be filled with *interstitial fluid*, which is basically water and all the molecules that are pushed out with it from the blood. (See [Chapter 11](#) for an explanation of how this happens.) In others, such as epithelial tissues, the function is dependent on the cells being tightly packed together. This is done by *cell junctions* which come in three general varieties:

- » **Tight Junctions:** Cells are basically sewn together using proteins leaving little to no space between them. These can be found in most epithelial

tissues.

- » **Desmosomes or Adhesive Junctions:** Proteins between the cells connect with filaments inside of them acting much like Velcro. Cells are still bound together tightly but have more flexibility as a layer. These can be found in the epidermis (the outer layer of skin) and the lining of the uterus.
- » **Gap Junctions:** Cells are connected by proteins that form channels, allowing the two neighboring cells to directly share cytoplasm. This junction enables the cells to pass on small molecules without investing energy. These can be found in most tissues of the body but are especially important in cardiac muscle.

In this chapter, you find a quick review of the four tissue types along with practice questions to test your knowledge of them.

Getting into Your Skin: Epithelial Tissue

Perhaps because of its unique job of lining all our body surfaces (both external and internal), epithelial tissue has many characteristics that distinguish it from other tissue types.

Epithelial tissues, which are arranged in sheets of tightly packed cells, always have a free, or *apical*, surface that is exposed to the air or to fluid. The cells have a defined top and bottom and can be found in a single layer or sandwiched between other cells of the same type. Cells on the apical side sometimes have cytoplasmic projections such as *cilia*, hair-like growths that can move material over the cell's surface, or *microvilli*, fingerlike projections that increase the cell's surface area for increased adsorption and more efficient diffusion of absorbents and secretions.



TIP

Adsorption involves the *adhesion* of something to a surface, while *absorption* involves a fluid permeating or dissolving into a liquid or solid.

Opposite the apical side is the *basal* side, or bottom layer of cells. These are attached to the tissue's *basement membrane* which is a layer of non-living material secreted by the epithelial cells. It adheres to the underlying connective tissue and acts as a filter — which is especially useful since epithelial tissues lack blood flow.

Epithelial tissue serves several key functions, including the following:

- » **Protection:** Skin protects vulnerable structures or tissues deeper in the body.
- » **Barrier:** Epithelial tissues prevent foreign materials from entering the body and retain interstitial fluids.
- » **Sensation:** Sensory nerve endings embedded in epithelial tissue connect the body with outside stimuli.
- » **Secretion:** Epithelial tissue in glands can be specialized to secrete enzymes, hormones, and fluids.
- » **Absorption:** Linings of the internal organs are specialized to facilitate absorption of materials into the body.
- » **Filtration:** The epithelium of the respiratory tract protects the body by filtering out dirt and particles and cleaning the air that's inhaled. The blood is filtered in epithelial tissue in the kidneys.
- » **Diffusion:** Simple squamous epithelial cells form a semipermeable membrane that allows diffusion of some materials to pass through under osmotic pressure. This way, molecules diffuse between the cells rather than through them.



REMEMBER Single-layer epithelial tissue is classified as *simple*. Tissue with more than one layer is called *stratified*. Epithelial tissues are also classified according to shape:

- » **Squamous:** *Squamous* cells are thin, flat, and odd-shaped.
- » **Cuboidal:** As the name implies, *cuboidal* cells are equal in height and width; shaped like cubes.

» **Columnar:** *Columnar* cells are taller than they are wide.



REMEMBER Following are the ten primary types of epithelial tissues:

- » **Simple squamous epithelium:** This flat layer of scale-like cells looks like a fish's scales and is useful in diffusion and filtration. Each cell nucleus is centrally located and is round or oval. Simple squamous epithelium lines the lungs' air sacs where oxygen and carbon dioxide are exchanged.
- » **Simple cuboidal epithelium:** These cube-shaped cells, found in a single layer that looks like a microscopic mattress, have centrally located nuclei that usually are round. They are found in the ovaries, line the tubes of the kidneys, and line some glands such as the thyroid, sweat glands, and salivary glands, as well as the inner surface of the eardrum, known as the *tympanic membrane*. Functions of this type of epithelium are secretion, absorption, and tube formation. The kidney tubules have microvilli that increase the area of adsorption for more efficient diffusion and absorption.
- » **Simple columnar epithelium:** These densely packed cells are taller than they are wide, with elongated nuclei located near the base of each cell. Found lining the digestive tract from the stomach to the anal canal, the functions of this type of epithelium are secretion and absorption.
- » **Ciliated simple columnar epithelium:** A close cousin to simple columnar epithelium, this type of tissue has hairlike cilia that can move mucus and other substances across the cell. It's found lining the small respiratory tubes.
- » **Pseudostratified columnar epithelium:** Pay attention to the prefix *pseudo*— here, which means “false.” It may look multilayered because the cells' nuclei are scattered at different levels, but it's not. This type of epithelium is found in the large ducts of the parotid glands (salivary glands) and some segments of the male reproductive system, including the urethra.
- » **Ciliated pseudostratified columnar epithelium:** Another variation on a theme, this tissue is nearly identical to pseudostratified columnar

epithelium. The difference is that the cells on this tissue's free surface have cilia, making it ideal for lining air passages because the cilia's uniform waving action causes dust and dirt particles trapped in a thin layer of mucus to move in one direction — away from the lungs and toward the throat and mouth. It's the insertion of these mucus-producing cells, called goblet cells, that cause the neighboring cells to lose their uniform, columnar shape.

- » **Stratified squamous epithelium:** This tissue is the stuff you see every day — your outer skin, or epidermis. Found in areas where the outer cell layer is constantly worn away, this type of epithelium regenerates its surface layer with cells from lower layers.



WARNING Though the newly produced cells along the basement membrane look more cuboidal, the cells in the other layers have the characteristic squamous shape. Don't fall into the trap of calling it stratified cuboidal!

- » **Stratified cuboidal epithelium:** This epithelium consists of two or three layers of cube shaped cells. It can be found in sweat glands and the male urethra. Its function is primarily protection of ducts in their neighboring tissue.
- » **Stratified columnar epithelium:** Also multilayered, this epithelium is found lining parts of the male urethra, excretory ducts of glands, the pharynx, and the conjunctiva of the eye. It has a protective function as well as secretion.
- » **Transitional epithelium:** This multilayered epithelium is referred to as *transitional* because its cells can shape-shift from cubes to squamous-like flat surfaces and back again. Found lining the urinary bladder, the cells stretch and flatten out (appearing squamous) to make room for urine. When urination occurs, the cells relax and assume their original form (appearing cuboidal).

Following are some practice questions dealing with epithelial tissue.



EXAMPLE

Q. Stratified epithelial tissue can be described as

- a. a thin sheet of cells.
- b. covered in cilia.
- c. layers of stacked epithelial cells.
- d. a long string of tissue.

A. The correct answer is layers of stacked epithelial cells. Remember that “stratified” means layers.

1-3 Match the epithelial tissue with its location in the body.

- a. desmosome
 - b. gap junction
 - c. tight junction
1. _____ Binds cells tightly, leaving little to no room between them
 2. _____ Tightly connects cells with a tunnel-like protein, allowing them to share their cytoplasmic contents
 3. _____ Adheres cells together by linking filaments of their cytoskeleton, allowing the tissue to better resist forces

4 Transitional epithelial tissue

- a. primarily protects specific areas.
- b. has the ability to stretch.
- c. is constantly worn away and replaced.
- d. is tailor-made for absorption.
- e. is found in the salivary glands.

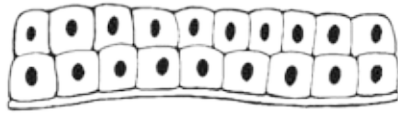
5-9 Match the epithelial tissue with its location in the body.

- a. Urinary bladder
 - b. Tubules of the kidney
 - c. Digestive tract
 - d. Epidermis of the skin
 - e. Respiratory passages
-
- 5. _____ Simple columnar
 - 6. _____ Stratified squamous
 - 7. _____ Transitional
 - 8. _____ Pseudostratified ciliated columnar
 - 9. _____ Simple cuboidal

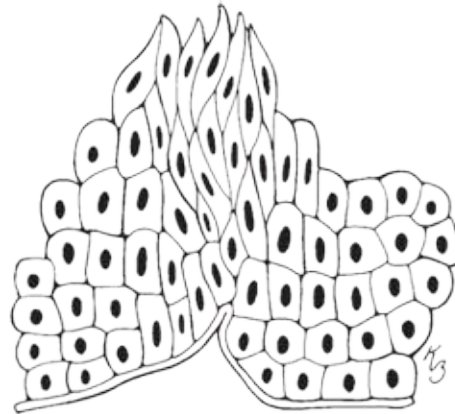
10 The key functions of epithelial tissue do **NOT** include

- a. protection.
- b. secretion.
- c. contraction.
- d. filtration
- e. absorption.

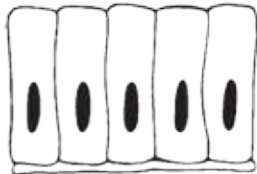
11-18 Identify the epithelial tissues shown in [Figure 5-1](#).



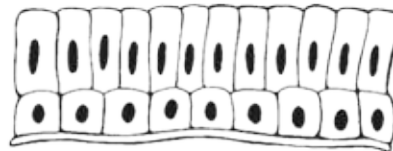
11. _____



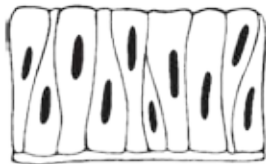
15. _____



12. _____



16. _____



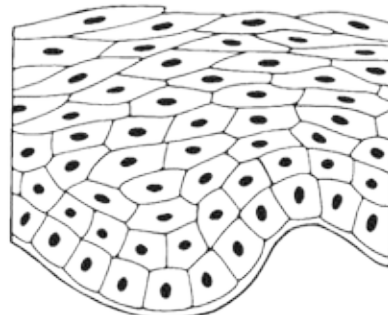
13. _____



17. _____



14. _____



18. _____

Illustration by Kathryn Born, MA

FIGURE 5-1: Epithelial tissues.

Making a Connection: Connective

Tissue

Connective tissues connect, support, and bind body structures together. Unlike other types of tissues, connective tissues are classified more by the stuff in which the cells lie — the extracellular *matrix* — than by the cells themselves. The matrix contains numerous fibrous proteins embedded in the *ground substance*, which is a thin gel. (It's often referred to as the extracellular fluid.) In most cases, the cells that produce that matrix are scattered within it. The load-bearing strength of connective tissue comes from a fibrous protein called *collagen*. All connective tissues contain a varying mix of collagen, elastic fibers for stretch, and reticular fibers which form a supportive mesh.



REMEMBER Following are the primary types of connective tissue:

- » **Areolar, or loose, tissue:** This tissue exists between and around almost everything in the body to bind structures together and fill space. It's made up of thick, wavy ribbons of *collagenous fibers*, and cylindrical threads called *elastic fibers*, which contain the protein elastin. Various cells, including lymphocytes, fibroblasts, fat cells, and mast cells, are scattered throughout the ground substance, leaving ample room for blood and lymphatic vessels as well as nerves (see [Figure 5-2](#)).
- » **Dense regular connective tissue:** Made up of parallel, densely packed bands or sheets of fibers (see [Figure 5-2](#)), this type of tissue is found in tendons, ligaments, surrounding joints, and anchoring organs.
Dense regular connective tissue has great tensile strength that resists lengthwise pulling forces. Ligaments contain more elastic fibers than tendons, making them more flexible. Neither have much blood flow though, which is why an injury to a tendon or ligament takes so long to heal.
- » **Dense irregular connective tissue:** This tissue has the same components as dense regular but appears much less organized. Its fibers twist and weave around each other, forming a thick tissue that can withstand stresses applied from any direction and leave space for other structures.

This tissue makes up the strong inner skin layer called the *dermis* as well as the outer capsule of organs like the kidney and the spleen.

- » **Adipose tissue:** Composed of fat cells, or *adipocytes*, this tissue forms padding around internal organs, provides insulation, and stores energy in fat molecules called triglycerides. Fat molecules fill the cells, forcing the nuclei against the cell membranes and giving them a bubble-like shape. Adipose has an *intracellular matrix* rather than an extracellular matrix.
- » **Reticular tissue:** Literally translated as “web-like” or “net-like,” reticular tissue is made up of slender, branching reticular fibers with reticular cells (a specialized fibroblast) overlaying them. Its intricate structure makes it a particularly good filter, which explains why it’s found inside the spleen, lymph nodes, and bone marrow.
- » **Cartilage:** These firm but flexible tissues, made up of collagen and elastic fibers, have no nerve cells or blood vessels (a state called *avascular*). Cartilage contains openings called *lacunae* (from the Latin word *lacus* for “lake” or “pit”) that enclose mature cells called *chondrocytes*. The ground substance is full of proteoglycans making it thicker than other connective tissues. There are three types of cartilage:
 - **Hyaline cartilage:** The most abundant cartilage in the body, it’s collagenous and made up of a uniform matrix pocked with chondrocytes. It lays the foundation for the embryonic skeleton, forms the costal (rib) cartilages, makes up nose cartilage, and covers the articulating surfaces of bones that form movable joints.
 - **Fibrocartilage:** As the name implies, fibrocartilage contains thick, compact collagen fibers. The sponge-like structure, with the lacunae and chondrocytes lined up within the fibers, makes it a good shock absorber. It’s found in the intervertebral discs of the vertebral column and in the symphysis pubis at the front of the pelvis.
 - **Elastic cartilage:** Similar to hyaline cartilage but with a much greater abundance of elastic fibers, elastic cartilage has more tightly packed lacunae and chondrocytes between parallel elastic fibers. This cartilage — which makes up structures where a specific form is important such as the outer ear and epiglottis — tends to bounce back to its original shape after being bent.

- » **Osseous, or bone, tissue:** Essentially, bone is mineralized connective tissue, making it the most rigid of the group. The characteristic collagen fibers are coated in mineral deposits giving it the hardness we associate with our bones. Cells called *osteoblasts* mineralize the matrix into repeating patterns called *osteons*. In the center of each osteon is a large opening, the *Haversian canal* (or *central canal*) that contains blood vessels, lymph vessels, and nerves. The central canal is surrounded by thin layers of bony matrix called *lamellae* that contain the lacunae, which house *osteocytes* (mature bone cells). Smaller tunnels called *canaliculi* connect the lacunae and the arms of the osteocytes form gap junctions within them. (We explore bone in more detail in [Chapter 7](#).)
- » **Blood:** Yes, blood is considered a type of connective tissue. Like other connective tissues, it has an extracellular matrix — in this case, plasma — in which *erythrocytes* (red blood cells), *leukocytes* (white blood cells), and *thrombocytes* (platelets) are suspended. (Blood also is considered a vascular tissue because it circulates inside arteries and veins, and we get into that in [Chapter 11](#).) Roughly half of blood's volume is fluid or plasma while the other half is suspended cells. Erythrocytes are concave on both sides and contain a pigment, *hemoglobin*, which carries oxygen throughout the body and takes carbon dioxide away. There are approximately 5 million erythrocytes per cubic millimeter of whole blood. Thrombocytes, which number approximately 250,000 per cubic millimeter, are fragments of cells used in blood clotting. Some leukocytes are large *phagocytic* cells (literally “cells that eat”) that are part of the body's immune system. There are, however, relatively few of them — less than 10,000 per cubic millimeter.

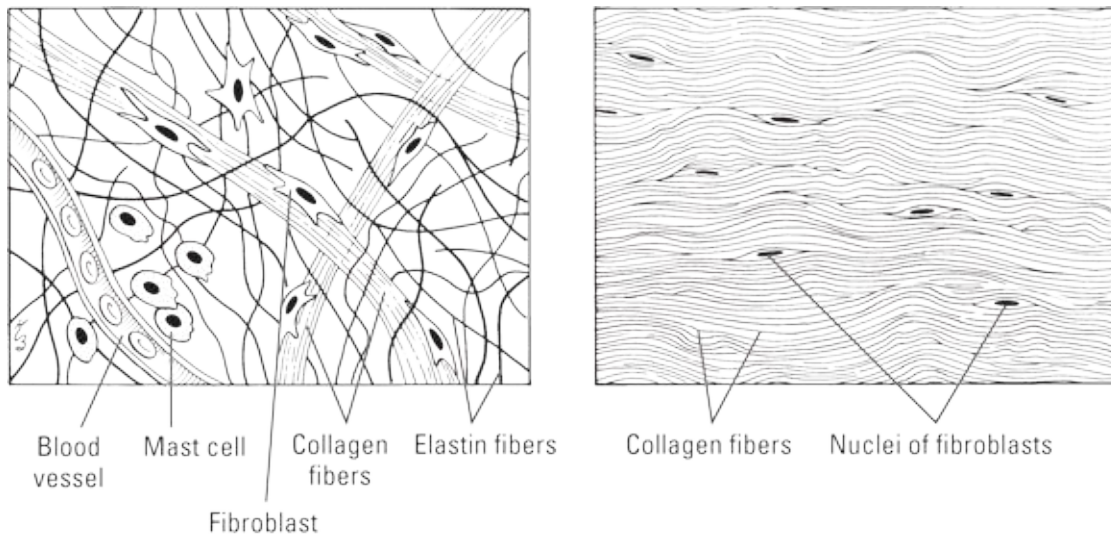


Illustration by Kathryn Born, MA

FIGURE 5-2: Areolar tissue and dense regular connective tissue.

19 What is adipose tissue composed of?

- a. Mast cells
- b. Chondrocytes
- c. Osteocytes
- d. Leukocytes
- e. Fat cells

20 What are tendons composed of?

- a. Elastic cartilage
- b. Dense regular connective tissue
- c. Areolar tissue
- d. Fibrocartilage
- e. Dense irregular connective tissue

21 Hyaline cartilage

- a. covers the surface of articulating bones.
- b. lines the symphysis pubis.

- c. forms the epiglottis.
- d. forms the outer ear and nose.
- e. cushions intervertebral discs.

22 Why does dense irregular connective tissue twist and weave around?

- a. To fill gaps between other tissues
- b. To withstand stresses applied from any direction
- c. To prevent dead or dying cells from weakening its structure
- d. To provide stretch and elasticity
- e. To provide firm connections between muscles and bones

23 Which tissue contains lacunae with osteocytes?

- a. Elastic cartilage
- b. Bone
- c. Hyaline cartilage
- d. Blood
- e. Reticular

24 What function do thrombocytes serve?

- a. They block invading microbes.
- b. They carry oxygen to connective tissues.
- c. They nourish osteocytes.
- d. They contribute to blood clotting.
- e. They provide nutrients to erythrocytes.

Flexing It: Muscle Tissue

Although we review how muscles work in [Chapter 8](#), in histology you should know that muscle tissue is made up of cells, which are referred to as fibers, known as *myocytes*. The cytoplasm within the fibers is called *sarcoplasm*,

and within that sarcoplasm are minute *myofibrils* that contain the protein filaments *actin* and *myosin*. These filaments slide past each other during a muscle contraction, shortening the fiber.



REMEMBER Following are the three types of muscle tissue (see [Figure 5-3](#)):

- » **Skeletal muscle tissue:** Biceps, triceps, pecs — these are the muscles that bodybuilders focus on. As the name implies, skeletal muscles attach to the skeleton and are controlled throughout by the conscious (voluntary) control function of the central nervous system for movement. Muscle fibers are cylindrical with several nuclei in each cell (which makes them *multinucleated*) and obvious *striations* (stripes that span the width of the cell) throughout. They are bundled in parallel and are roughly the length of the muscle they make up.
- » **Cardiac muscle tissue:** This tissue is composed of cylindrical fibers, *cardiomyocytes*, each with a central nucleus and are striated. The tissue forms a branching pathway as one cell can join with two or more fibers at one end using gap junctions called *intercalated discs*. Cardiac muscle tissue contractions occur through the autonomic nervous system (involuntary control).
- » **Smooth muscle tissue:** Made up of spindle-shaped fibers with large, centrally located nuclei, it's found in the walls of internal organs, or *viscera*. As with cardiac muscle, this type of tissue contracts without conscious control. Smooth muscle gets its name from the fact that, unlike other muscle tissue types, it is not striated.

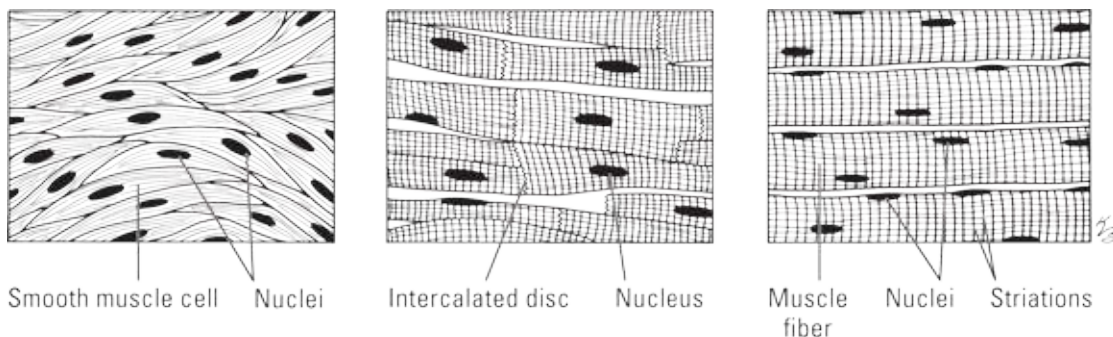


Illustration by Kathryn Born, MA

FIGURE 5-3: Muscle tissues: Smooth, cardiac, and skeletal.

25 Which type of tissue is multinucleated?

- a. Skeletal muscle tissue
- b. Cardiac muscle tissue
- c. Smooth muscle tissue
- d. Autonomic muscle tissue

26 Both skeletal and cardiac muscle tissue have prominent lines across the fiber. What are they called?

- a. Fibroblasts
- b. Multinucleates
- c. Lacunae
- d. Striations
- e. Intercalated discs

27 Which of the following contains smooth muscle tissue?

- a. The heart
- b. The urinary bladder
- c. The biceps
- d. The deltoid

28 Describe the location and function of intercalated discs.

Getting the Signal Across: Nervous Tissue

There's only one type of nervous tissue and only one primary type of cell in it: the *neuron*. There are numerous other cells, called *glial cells*, but they don't perform the communication function we associate with nervous tissue. A neuron is unique in that it can both generate and conduct electrical signals in the body. That process starts when sense receptors receive a stimulus that causes an electrical signal to be sent through cytoplasmic projections called *dendrites*. From there, the signal moves through the *cell body* to the *axon*, another cytoplasmic extension. Here, an impulse is generated and spreads to the end of the axon, where it hands the signal off chemically to the next cell down the line. (We look more closely at how all that happens in [Chapter 9](#).)

- 29** True or False: Nervous tissue is essentially formed by neurons connecting end-to-end with very little matrix.
- 30** Which statement best describes how neurons communicate?
- a. Axons generate impulse, which is passed to dendrites through the cell body.
 - b. Axons receive a stimulus, which is passed to the cell body, triggering the dendrites to release a chemical.
 - c. Cell bodies respond to a stimulus by generating an electrical signal to pass on to the dendrites and axons, where the former will pass it on electrically and the latter will do so chemically.
 - d. In response to stimulus, the dendrite generates an electrical signal that is passed to the cell body. The axon will then spread an impulse down its length, releasing a chemical at the end.
 - e. Dendrites respond to stimulus by releasing a chemical within the cell that travels through the cell body. The axon generates an impulse in response, spreading it to the end, where it will pass the signal on to the next cell.

Answers to Questions on Histology

The following are answers to the practice questions presented in this chapter.

- 1-3 1. **c. tight junction**; 2. **b. gap junction**; 3. **a. desmosome**
- 4 Transitional epithelial tissue **b. has the ability to stretch**. This tissue lines the bladder, so it had better be stretchy!
- 5 Simple columnar: **c. Digestive tract**
- 6 Stratified squamous: **d. Epidermis of the skin**
- 7 Transitional: **a. Urinary bladder**
- 8 Pseudostratified ciliated columnar: **e. Respiratory passages**
- 9 Simple cuboidal: **b. Tubules of the kidney**
- 10 The key functions of epithelial tissue do *not* include **c. contraction**. Muscles contract; epithelial tissue does not.
- 11-18 Following is how [Figure 5-1](#), the types of epithelial cells and tissues, should be labeled.

11. **stratified cuboidal**; 12. **simple columnar** 13. **pseudostratified**; 14. **simple squamous**; 15. **transitional**; 16. **stratified columnar**; 17. **simple cuboidal**; 18. **stratified squamous**



WARNING Identifying epithelials can be a bit tricky. First look for the basement membrane and the space then look only at the cells in between. Simple cuboidal can appear stratified because the tissue forms tubes and neighboring tubes will connect at their basement membrane. When stratified, you must look at the pattern of the cells throughout. Stratified squamous has cube-shaped cells along the bottom but the bulk of the cells are clearly squamous. Transitional is the most troublesome but you will see both squamous and cuboidal throughout the width of the tissue.

- 19 Adipose tissue is composed of **e. fat cells**. Think of the Latin *adepts*, which means “fat.”

- 20) Tendons are composed of **b. dense regular connective tissue**. Tendons are dense and exhibit a regular pattern.
- 21) Hyaline cartilage **a. covers the surface of articulating bones**. It's all in the names of the three types of cartilage. Hyaline comes from the Greek word for *glass*, which seems appropriate for something covering a surface. Fibrocartilage contains thick bunches of fibers, which doesn't sound right for covering a surface. And elastic cartilage tends to be found where exact shapes are important, such as in the outer ear.
- 22) Why does dense irregular connective tissue twist and weave around? **b. To withstand stresses applied from any direction**. The irregular pattern also creates space for structure like blood vessels to pass through.
- 23) Which tissue contains lacunae with osteocytes? **b. Bone**. Remember that *osteo* is the Latin word for "bone."
- 24) What function do thrombocytes serve? **d. They contribute to blood clotting**. Knowing that the Greek word *thrombos* means "clot" can help you spot the correct answer in this question.
- 25) Which type of tissue is multinucleated? **a. Skeletal muscle tissue**. Cardiac muscle tissue and smooth muscle tissue typically have only one nucleus per cell; autonomic muscle tissue is a trick answer choice.
- 26) Both skeletal and cardiac muscle tissue have prominent lines across the fiber. What are they called? **d. Striations**. Striations — think *stripes* — are alternating light and dark lines.
- 27) Which of the following contains smooth muscle tissue? **b. The urinary bladder**. The other answer choices contain striated tissue, which technically means that they aren't smooth.
- 28) Describe the location and function of intercalated discs. **Intercalated discs are found where cardiac muscle fibers meet end-to-end. These gap junctions form tunnels that allow cells to pass molecules to each other directly, without expending energy on endo- and exocytosis.**
- 29) Nervous tissue is essentially formed by neurons connecting end-to-end

with very little matrix. **False:** Abundant space is visible between neurons dotted with glial cells and numerous “fibers,” which are actually the ends of axons belonging to other neurons.

- 30) Which statement best describes how neurons communicate? **d. In response to stimulus, the dendrite generates an electrical signal that is passed to the cell body. The axon will then spread an impulse down its length, releasing a chemical at the end.** While neurons communicate with themselves using electrical signaling, that cannot be passed to another cell so chemicals must be used. While both dendrites and axons can generate an electric signal, it is not termed impulse until the axon does it.

Part 2

Weaving It Together: Bones, Muscles, and Skin

IN THIS PART ...

Wrap the package in the body's largest single organ: the skin. Get the scoop on the epidermis and the dermis; find out about nerves in the skin; and accessorize with hair, nails, and glands.

Build on a strong foundation: Explore how bones are formed. Piece those bones together into a skeleton and distinguish your axial from your appendicular.

Articulate some joints and attach muscles to that framework. Contract and relax to leverage muscle power.

Chapter 6

It's Skin Deep: The Integumentary System

IN THIS CHAPTER

- » Going below the surface of the skin
 - » Getting on your nerves
 - » Checking out hair, nails, and glands
-

Did you know that the skin is the body's largest organ? In an average person, its 17 to 20 square feet of surface area represents 5-7 percent of the body's weight. Self-repairing and surprisingly durable, the skin is the first line of defense against the harmful effects of the outside world. It helps retain moisture; regulates body temperature; hosts the sense receptors for touch, pain, and heat; and excretes excess salts and small amounts of waste.

Skin is jam-packed with components; it has been estimated that every square inch of skin contains 15 feet of blood vessels, 4 yards of nerves, 650 sweat glands, 100 oil glands, 1,500 sensory receptors, and more than 3 million cells with an average life span of 26 days that are constantly being replaced.

In this chapter, we peel back the surface of this most-visible organ system. We also give you plenty of opportunities to test your knowledge.

Digging Deep into Dermatology

Skin — together with hair, nails, and glands — composes the *integumentary system* (shown in [Figure 6-1](#)). The name stems from the Latin verb *integere*, which means “to cover.” The relevant Greek and Latin roots include *dermato* and *cutis*, both of which mean “skin.”

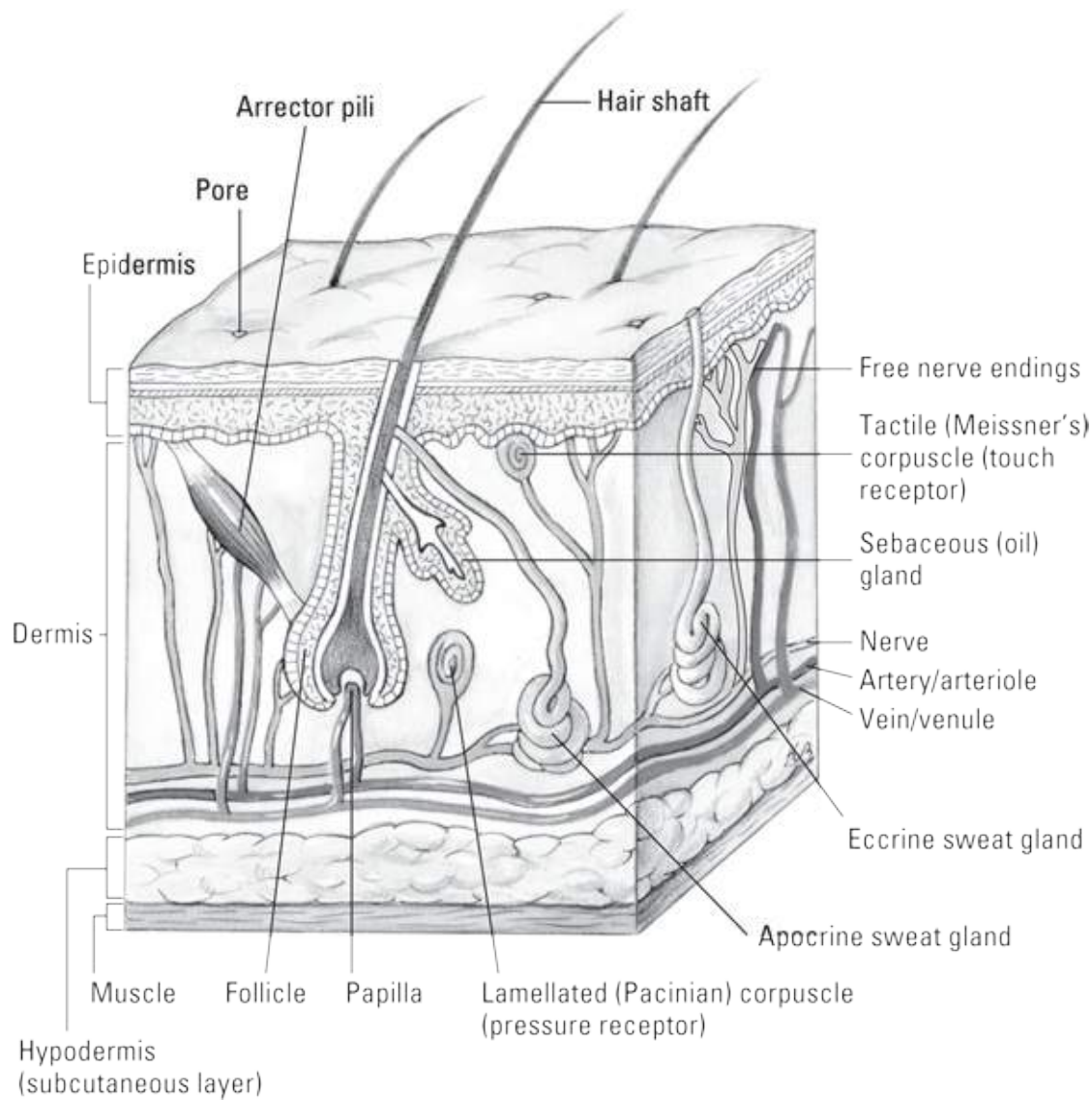


Illustration by Kathryn Born, MA

FIGURE 6-1: The integumentary system.

The skin consists of two primary parts, which we describe in the following sections: the *epidermis* and the *dermis*. (The Greek root *epi-* means “upon” or “above.”) Underlying the epidermis and dermis is the *hypodermis* or *superficial fascia* (also sometimes called *subcutaneous tissue*), which acts as a foundation but is not technically part of the skin. Composed of areolar and adipose tissue, it anchors the skin through fibers that extend from the dermis. Underneath, it attaches loosely to tissues and organs so that muscles can move freely. The fat in the hypodermis buffers deeper tissues and acts as insulation, preventing heat loss from within the body’s core. The dermis and the hypodermis are home to pressure-sensitive nerve endings called

lamellated or *Pacinian corpuscles* that respond to a deeper poke in the skin.

The epidermis: Don't judge this book by its cover



REMEMBER The epidermis, which contains no blood vessels, is made up of layers of closely packed epithelial cells. From the outside in, these layers are the following:

- » **Stratum corneum** (literally the “horny layer”) is about 20 layers of flat, scaly, dead cells containing a type of water-repellent protein called *keratin*. These cells, which represent about three-quarters of the thickness of the epidermis, are said to be *cornified*, which means that they’re tough and horny like the cells that form hair or fingernails. Humans shed this layer of tough, durable skin at a prodigious rate. Where the skin is rubbed or pressed more often, cell division increases, resulting in calluses and corns.
- » **Stratum lucidum** (from the Latin word for “clear”) is found only in the thick skin on the palms of the hands and the soles of the feet. This translucent layer of dead cells contains *eleidin*, a protein that becomes keratin as the cells migrate into the stratum corneum, and it consists of cells that have lost their nuclei and cytoplasm, hence they appear clear.
- » **Stratum granulosum** is three to five layers of flattened cells containing *keratohyalin*, a substance that marks the beginning of keratin formation. No nourishment from blood vessels reaches this far into the epidermis, so cells are either dead or dying by the time they reach the stratum granulosum. The nuclei of cells found in this layer are degenerating; when the nuclei break down entirely, the cell can’t metabolize nutrients and dies.
- » **Stratum spinosum** (also sometimes called the *spinous layer*) has five to ten layers containing *spinous*, or *prickle cells*, named for the spinelike projections that connect them with other cells in the layer. *Langerhans cells*, which are involved in the body’s immune response, are prevalent in the upper portions of this layer and sometimes the lower part of the stratum granulosum; they migrate from the skin to the lymph nodes in response to infection.

» **Stratum basale** (or *stratum germinativum*) is also referred to as the *germinal layer* because this single layer is mostly composed of cube-shaped stem cells. This is the site of mitosis as these stem cells divide and push cells upward. It takes about two weeks for the cells that originate here to migrate up to the stratum corneum, and it's another two weeks before they're shed. It rests on the papillary (rough or bumpy) surface of the dermis, close to the blood supply needed for nourishment and oxygen. About a quarter of this layer's cells are *melanocytes*, cells that synthesize a pale yellow to black pigment called *melanin* that contributes to skin color and provides protection against ultraviolet radiation (the kind of radiation found in sunlight). The remaining basal cells divide into *keratinocytes*, the primary epithelial cell of the skin. Melanocytes secrete melanin and the keratinocytes bring it in via a process called *cytokrine secretion*, thus pigmenting the cells throughout the epithelium. *Merkel cells*, large nervous cells involved in the sense of touch, occasionally appear amid the keratinocytes.

In addition to melanin, the epidermis contains a yellowish pigment called *carotene* (the same one found in carrots and sweet potatoes). Found in the stratum corneum and the fatty layers beneath the skin, it produces the yellowish hue associated with Asian ancestry or increased carrot consumption. The pink to red color of Caucasian skin is caused by *hemoglobin*, the red pigment of the blood cells. Because Caucasian skin contains relatively less melanin, hemoglobin can be seen more easily through the epidermis. Sometimes the limited melanin in Caucasian skin pools in small patches. Can you guess the name of those patches of color? Yep, they're freckles. Albinos, on the other hand, have no melanin in their skin at all, making them particularly sensitive to ultraviolet radiation.

Deep to the stratum basale is the *basement membrane* of the epidermal tissue. It is thought to be created by the layer itself. In addition to anchoring the epidermal layer to the dermis below. It also acts as a filter — controlling what gets up into this layer from below and what is allowed deeper.

Uplifted portions of the dermis, *dermal papillae* (fingerlike projections), cause ridges and grooves to form on the outer surface of the epidermis that increase the friction needed to grasp objects or move across slick surfaces. On hands and feet, these ridges form patterns of loops and whorls — fingerprints,

palm prints, and footprints — that are unique to each person. You leave these imprints on smooth surfaces because of the oily secretions of the oil and sweat glands on the skin's surface.

The dermis: Going more than skin deep



REMEMBER Beneath the epidermis is a thicker, fibrous structure called the dermis. It consists of the following two layers, which blend together:

- » **The outer, soft *papillary layer*** contains areolar tissue that projects into the epidermis to bring blood and nerve endings closer. The papillae increase the surface area of the connection between the dermis and epidermis' basement membrane. Some of these papillae contain *Meissner's corpuscles*, nervous receptors that are sensitive to soft touch.
- » **The inner, thicker *reticular layer*** (from the Latin word *rete* for “net”) is made up of dense irregular connective tissue containing interlacing bundles of collagenous and elastic fibers that form the strong, resistant layer used to make leather and suede from animal hides. This layer is what gives skin its strength, extensibility, and elasticity. Within the reticular layer are *sebaceous glands* (oil glands), sweat glands, fat cells, hair follicles, nerve cells, and larger blood vessels.

The dermis also includes *fibroblasts* (which create the fibers that form the bulk of connective tissues), and *macrophages* (which engulf waste and foreign microorganisms). Thickest on the palms of the hands and soles of the feet, the dermis is thinnest over the eyelids, penis, and scrotum. It's thicker on the back (posterior) than on the stomach (anterior) and thicker on the sides (lateral) than toward the middle of the body (medial). The various skin “accessories” — blood vessels, nerves, glands, and hair follicles — are embedded here. Adipocytes (fat cells) are found at the bottom of the dermis, which transitions into the hypodermis below.

See if you've got the skinny on skin so far:



EXAMPLE Q. The layer of epidermis that's composed of a horny, cornified tissue that sloughs off is called the stratum

- a. corneum.
- b. lucidum.
- c. granulosum.
- d. spinosum.
- e. basale.

A. The correct answer is corneum. Cornified, corns — think of how hard a kernel of popcorn can be.

1 What does the papillary layer of the dermis *not* do?

- a. Filter out microbes
- b. Extend into the epidermis
- c. Carry the blood and nerve endings close to the epidermis
- d. Aid in holding the epidermis and dermis together
- e. Contain cells sensitive to touch

2 The epidermal ridges on the fingers function to

- a. provide a means of identification.
- b. increase the friction of the epidermal surface.
- c. decrease water loss by the tissues.
- d. aid in regulating body temperature.
- e. prevent bacterial infection.

3 Caucasian skin color is caused by

- a. carotene pigment in the dermis.
- b. the high level of melanin in the epidermis.

- c. less melanin in the skin, allowing the blood pigment to be seen.
- d. the absence of all pigment.
- e. melanin and carotene pigments.

4 The sequence of layers in the epidermis from the dermis outward is

- a. corneum, lucidum, granulosum, spinosum, basale.
- b. corneum, granulosum, lucidum, basale, spinosum.
- c. spinosum, basale, granulosum, corneum, lucidum.
- d. basale, spinosum, granulosum, lucidum, corneum.
- e. basale, lucidum, corneum, spinosum, granulosum.

5 What's another name for the subcutaneous layer of tissue?

- a. Epidermis
- b. Superficial fascia
- c. Papillary layer
- d. Inner reticular layer
- e. Dermis

6 What do carrots and sweet potatoes have in common with human epidermis?

- a. They synthesize melanocytes.
- b. They have layers of cells at varying stages of development.
- c. They contain the pigment carotene.
- d. They secrete water proofing oils.
- e. They have a papillary surface.

7 What is the name of a layer of dense irregular connective tissue containing interlacing bundles of collagenous and elastic fibers?

- a. Basale layer of the epidermis

- b. Reticular layer of the dermis
- c. Outer layer of the hypodermis
- d. Papillary layer of the dermis
- e. Inner layer of the hypodermis

8 Why is keratin important to the skin?

- a. It makes the stratum corneum thick, tough, and water-repellant.
- b. It keeps the stratum lucidum moisturized.
- c. It helps nourish the stratum granulosum.
- d. It maintains connections between the cells of the stratum spinosum.
- e. It distributes sweat evenly through the epidermal layer.

9 The stratum _____ epidermal layer contains keratohyalin.

- a. germinativum
- b. spinosum
- c. lucidum
- d. granulosum
- e. corneum

10 What do the prickle cells in the spinous layer do?

- a. They metabolize nutrients.
- b. They connect the cells of the layer.
- c. They transform other cells that have lost their nuclei and cytoplasm.
- d. They move nutrients from the blood into the epidermis.
- e. They secrete keratohyalin.

11 What is the name for melanin that forms into patches?

- a. Flexion creases
- b. Freckles

- c. Dermal papillae
- d. Lamellated corpuscles
- e. Matrix

12 Briefly outline the steps of epidermal growth.

Touching a Nerve in the Integumentary System

At least four kinds of receptors are involved in creating the sensation of touch:

» **Free nerve endings:** These are the most abundant type of sensory endings, occurring widely in the integument and within muscles, joints, viscera, and other structures. Nerve endings are *dendrites* (branched extensions) of sensory neurons with a swelling at the end. These mostly function in *nociception*, or pain sensation, though some sense temperature, touch, and stretch. Found all over the body, free nerve endings are especially prevalent in epithelial and connective tissue.

The free nerve endings that sense temperature come in two varieties: ones that respond to cold stimuli and different ones that respond to heat. A warm sensation comes from the hypothalamus of the brain interpreting information from both.

» **Meissner's corpuscles:** These light-touch mechanoreceptors lie within the dermal papillae (you can see one in [Figure 6-1](#)). They're small, egg-shaped capsules of connective tissue surrounding a spiraled end of a

dendrite. They are the most abundant in sensitive skin areas such as the lips and fingertips.

- » **Pacinian corpuscles:** These deep-pressure mechanoreceptors are dendrites surrounded by concentric layers of connective tissue (check one out in [Figure 6-1](#)). Found deep within the dermis and hypodermis, they respond to deep or firm pressure and vibrations. Each is more than 2 millimeters long and therefore visible to the naked eye.
- » **Hair plexus:** These nerve endings wrap around the hair follicle and function as mechanoreceptors. When the hair is moved, it stimulates the nerve endings, creating a very light touch sensation.

Test whether you're staying in touch with this section:

13 Meissner's corpuscles play a role in which function?

- a. The sensation of heavy pressure
- b. The skin's ability to rebound into shape after pressure is applied
- c. The immediate withdrawal response from intense heat
- d. The sense of motion
- e. The sensation of soft touch

14 Which receptor type responds to a painful stimulus?

- a. Hair plexus
- b. Meissner's corpuscle
- c. Pacinian corpuscle
- d. Thermoreceptor
- e. Nociceptor

Accessorizing with Hair, Nails, and Glands

Mother Nature has accessorized your fashionable over-wrap with a variety of

specialized structures that grow from the epidermis: hair, fingernails, toenails, and several types of glands.

Wigging out about hair

Like most mammals, hair covers the entire human body except for the lips, eyelids, palms of the hands, soles of the feet, nipples, and portions of external reproductive organs. But human body hair generally is sparser and much lighter in color than that sported by most other mammals. Animals have hair for protection and temperature control. For humans, however, body hair is largely a secondary sex characteristic.

A thick head of hair protects the scalp from exposure to the sun's harmful rays and limits heat loss. Eyelashes block sunlight and deflect debris from the eyes. Hair in the nose and ears prevents airborne particles and insects from entering. Touch receptors connected to hair follicles respond to the lightest brush.

The average adult has about 5 million hairs, with about 100,000 of those growing from the scalp. Normal hair loss from an adult scalp is anywhere from 40 to 100 hairs each day, although baldness can result from genetic factors, hormonal imbalances, scalp injuries, disease, dietary deficiencies, radiation, or chemotherapy.

Each hair grows at an angle from a follicle embedded in the epidermis and extending into the dermis (refer to [Figure 6-1](#)); scalp hairs sometimes reach as far as the hypodermis. Nerves reach the hair at the follicle's expanded base, called the *bulb*, where a nipple-shaped papilla of connective tissue and capillaries provide nutrients to the growing hair. Epithelial cells in the bulb divide to produce the hair's *root*. Like the growth of epidermis, cells die as they are pushed away from the source of nutrients. By the time the hair reaches the scalp's surface, now referred to as the *shaft*, the cells have long been dead. The shape of a hair's cross section can vary from round to oval or even flat — influenced by the follicle's shape. Oval hairs grow out appearing wavy or curly, flat hairs appear kinky, and round hairs grow out straight. Each scalp hair grows for two to three years at a rate of about 1/3 to 1/2 millimeter per day, or 10 to 18 centimeters per year. When mature, the hair rests for three or four months before slowly losing its attachment. Eventually, it falls out and is replaced by a new hair.

Hair pigment (which is mostly melanin, just as in the skin) is produced by melanocytes in the follicle and transferred to the cells. Three types of melanin — black, brown, and yellow — combine in different quantities for each individual to produce different hair colors ranging from light blonde to black. Gray and white hairs grow in when melanin levels decrease and air pockets form where the pigment used to be.

Wondering why you have to shampoo so often? Hair becomes oily over time thanks to *sebum*, a mixture of cholesterol, fats, and other substances secreted from a *sebaceous* (or *holocrine*) gland found next to each follicle (see one for yourself in [Figure 6-1](#)). Sebum keeps both hair and skin soft, pliable, and waterproof. Attached to each follicle is a smooth muscle called an *arrector pili* (literally “raised hair”) that straightens the hair shaft, elevating the skin in a pattern called *goose bumps* or *goose pimples*. In other mammals this traps heat, keeping them warm. We, however, do not have enough hair for that benefit, but we certainly still get them!

The oil slick

The dermis is also home to our *sebaceous glands*, which are most often associated with a hair follicle (see [Figure 6-1](#)). They secrete a fatty substance called *sebum* which you know as oil. Since the sebaceous gland is attached to the follicle, when the arrector pili contract, the pressure put on the gland can cause it to release its sebum. While we find the accumulation of this oil to be annoying, especially since it’s the main cause of acne, it does have important functions. It softens the hairs and skin surface, making them both more pliable. It also serves to reduce friction as well as provide waterproofing.

Nailing the fingers and toes

The hard part of the fingernails and toenails is also made of keratin. Human nails (which actually are vestigial claws) have three parts: a *root bed* at the nail base, a *body* that’s attached to the fingertip, and a *free edge* that grows beyond the end of the finger or toe.

Heavily cornified tissue forms the nails from modified strata corneum and lucidum (we describe these and other layers in the earlier section “[The epidermis: Don’t judge this book by its cover](#)”). *Cornification* is the process by which squamous cells are basically turned into keratin. A narrow fold of the stratum corneum turns back to form the *eponychium*, or *cuticle*. Under the

nail, the nail bed is formed by the strata basale and spinosum. At the base of the nail, partially tucked under the cuticle, the strata thicken to form a whitish area called the *lunula* (literally “little moon”) that can be seen through the nail. Beneath the lunula is the *nail matrix*, a region of thickened strata where mitosis pushes previously formed cornified cells forward, making the nail grow. Under the free edge of the nail, the stratum corneum thickens to form the *hyponychium*. Nails are pinkish in color because of hemoglobin in the blood of the underlying capillaries, which are visible through the translucent cells of the nail.

On average, fingernails grow about 1 millimeter each week. Toenails tend to grow even more slowly. Nails function as an aid to grasping, as a tool for manipulating small objects, and as protection against trauma to the ends of fingers and toes.

Sweating the details

Humans perspire over nearly every inch of skin, but anyone with sweaty palms or smelly feet can attest to the fact that sweat glands are most numerous in the palms and soles, with the forehead running a close third. There are two types of sweat, or *sudoriferous*, glands: *eccrine* and *apocrine*. Both are coiled tubules embedded in the dermis and are composed of simple columnar cells (refer to [Figure 6-1](#)).

- » **Eccrine sweat glands** are distributed widely over the body — an average adult has roughly 3 million of them — and produce the watery, salty secretion you know as sweat. Each gland’s duct passes through the epidermis to the skin’s surface, where it opens as a *sweat pore*. The sympathetic division of the autonomic nervous system (see [Chapter 9](#)) controls when and how much perspiration is secreted depending on how hot the body becomes. Sweat helps cool the skin’s surface by evaporating as fast as it forms. About 99 percent of eccrine-type sweat is water, but the remaining 1 percent is a mixture of sodium chloride and other salts, uric acid, urea, amino acids, ammonia, sugar, lactic acid, and ascorbic acid.
- » **Apocrine sweat glands** are located primarily in armpits (known as the *axillary region*) and the external genital area. Rather than secreting onto the surface of the skin, these release sweat into the hair follicle. These glands don’t become active until puberty and are triggered by stress.

Although apocrine-type sweat contains the same basic components as eccrine sweat and is also odorless when first secreted, bacteria residing in the hair follicle quickly begin to break down its additional fatty acids and proteins — explaining the post-exercise underarm stench. In addition to exercise, sexual and other emotional stimuli can cause contraction of cells around these glands, releasing sweat.

Getting an earful

The occasionally troublesome yellowish substance known as earwax is secreted in the outer part of the ear canal from modified sudoriferous glands called *ceruminous glands* (the Latin word *cera* means “wax”). Lying within the subcutaneous layer of the ear canal, these glands have ducts that either open directly into the ear canal or empty into the ducts of nearby sebaceous glands, mixing their secretion with sebum and dead epithelial cells.

Technically called *cerumen*, earwax is the combined secretion of these two glands. Working with ear hairs, cerumen traps any foreign particles before they reach the eardrum. As the cerumen dries, it flakes and falls from the ear, carrying particles out of the ear canal.

Think you’ve got a grip on everything to do with hair, nails, and glands? Find out by answering the following practice questions:

15 What is another name for the cuticle?

- a. Lunula
- b. Hyponychium
- c. Eponychium
- d. Nail matrix
- e. Perionychium

16 The _____ glands form perspiration.

- a. sebaceous
- b. ceruminous
- c. endocrine
- d. Merkel

e. sudoriferous

17 The cause of graying hair is

- a. production of melanin in the shaft of the hair.
- b. production of carotene in the shaft of the hair.
- c. decrease in blood supply to the hair.
- d. lack of melanin in the shaft of the hair.
- e. parenthood.

18 From where does the hair develop?

- a. Arrector pili
- b. Shaft
- c. Follicle
- d. Sebaceous gland
- e. Lanugo

19 The nails are modifications of the epidermal layers

- a. corneum and lucidum.
- b. lucidum and granulosum.
- c. granulosum and spinosum.
- d. spinosum and basale.
- e. lucidum and spinosum.

20 Which factor is not associated with baldness?

- a. Genetics
- b. Hormonal imbalances
- c. Scalp injuries
- d. Lack of melanin
- e. Disease

21 Sebaceous glands

- a. produce a watery solution called sweat.
- b. produce an oily mixture of cholesterol, fats, and other substances.
- c. produce a waxy secretion called cerumen.
- d. accelerate aging.
- e. are associated with endocrine glands.

22 What function does the bulb at the base of a hair follicle serve?

- a. To prevent dirt and debris from becoming embedded in the skin
- b. To establish additional thermal protection
- c. To provide nutrients to the growing hair
- d. To regulate sweat production
- e. To inject melanin into the hair

23 This gland secretes water with fatty acids and proteins in response to stress. It acquires an unpleasant odor when bacteria breaks down the organic molecules it secretes.

- a. Apocrine sweat gland
- b. Sebaceous gland
- c. Ceruminous gland
- d. Eccrine sweat gland
- e. Mammary gland

24-27 Use the terms that follow to identify the structures of the nail shown in [Figure 6-2](#).

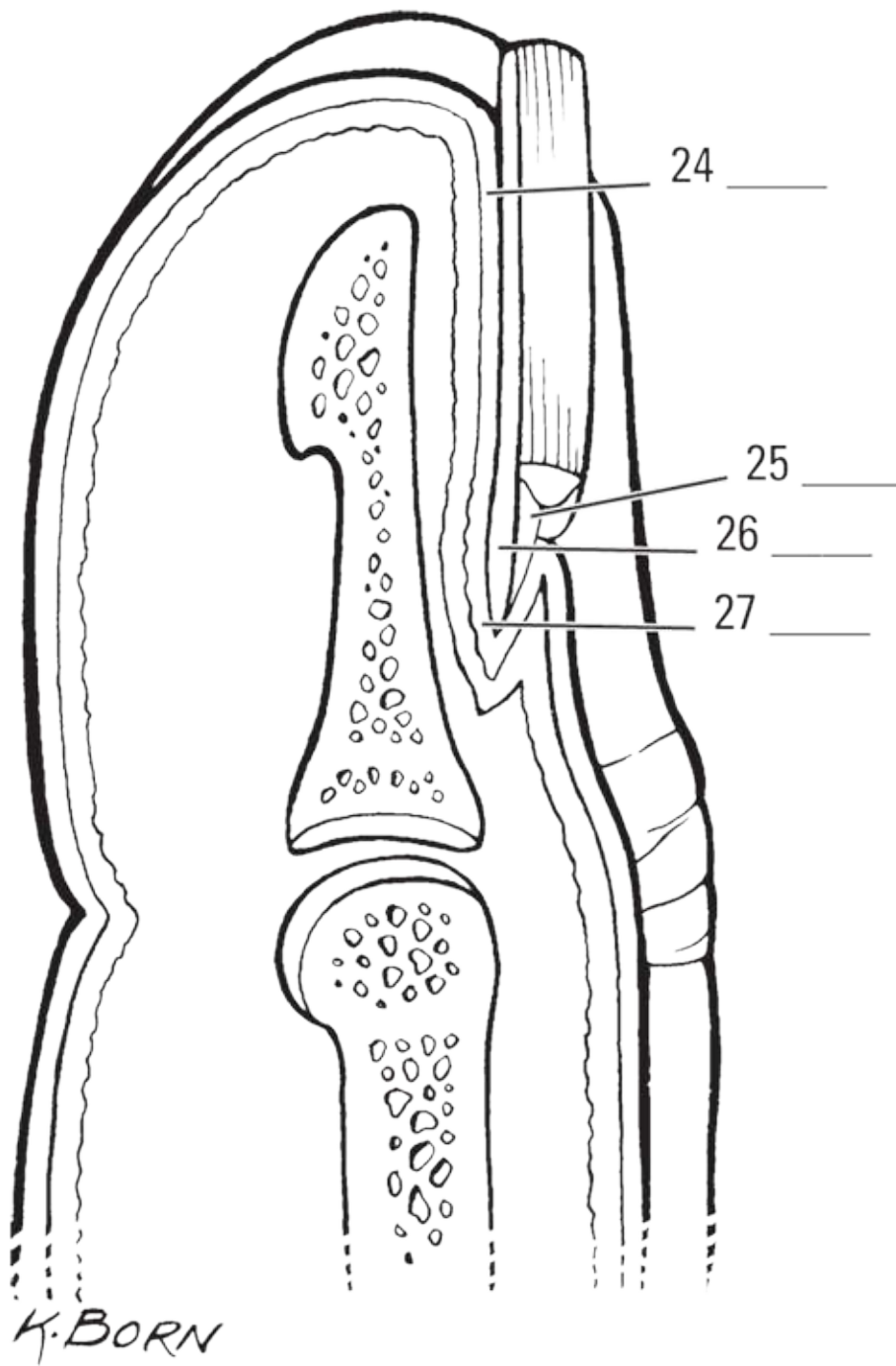


Illustration by Kathryn Born, MA

FIGURE 6-2: The fingernail.

- a. cuticle
- b. nail bed
- c. matrix

d. root

28 When the arrector pili contracts, which gland could incidentally release its contents?

- a. Apocrine sweat gland
- b. Eccrine sweat gland
- c. Ceruminous gland
- d. Sebaceous gland
- e. Endocrine gland

Answers to Questions on the Skin

The following are answers to the practice questions presented in this chapter.

- ① What does the papillary layer of the dermis not do? **a. Filter out microbes.** Busy little fingerlike projections, those papillae, but they're not filters. That function is carried out by the basement membrane of the epidermis.
- ② The epidermal ridges on the fingers function to **b. increase the friction of the epidermal surface.** They're Mother Nature's way of helping you cling to tree branches or grab food. While Choice a seems reasonable, that's a perk not a function.
- ③ Caucasian skin color is caused by **c. less melanin in the skin, allowing the blood pigment to be seen.** Here's a fun experiment: Turn off the lights, press your fingers together, and hold a flashlight under them. See the red glow? That's hemoglobin, too.
- ④ The sequence of layers in the epidermis from the dermis outward is **d. basale, spinosum, granulosum, lucidum, corneum.**



TIP Memory tool time: Base, spine, grain, Lucy, corny. Or try the first letters of Be Super Greedy, Less Caring. Insensitive, yes, but effective.

- 5) What's another name for the subcutaneous layer of tissue? **b. Superficial fascia.** Subcutaneous is the same as hypodermis (from the Greek *hypo*– for “beneath”).
- 6) What do carrots and sweet potatoes have in common with human epidermis? **c. They contain the pigment carotene.** Most of the other answers wouldn't apply to both mammalian and vegetable cells.
- 7) What is the name of a layer of dense irregular connective tissue containing interlacing bundles of collagenous and elastic fibers? **b. Reticular layer of the dermis.** The description in this question sounds like a tough structure, so it may help you to remember that the reticular layer is what's used to make leather from animal hides.
- 8) Why is keratin important to the skin? **a. It makes the stratum corneum thick, tough, and water-repellant.** Associate the words “corneum” and “keratin,” and you're in great shape.
- 9) The stratum **d. granulosum** epidermal layer contains keratohyalin. Keratohyalin eventually becomes keratin, so think of the layer where the cells are starting to die off.
- 10) What do the prickly cells in the spinous layer do? **b. They connect the cells of the layer.** The spine-like projections that make those connections also make the cells look prickly, hence the name. Don't get distracted by this layer being moisturized. The keratin doesn't do this; the sebum does.
- 11) What is the name for melanin that forms into patches? **b. Freckles.** Ever noticed how kids have more freckles at the end of a long summer spent outdoors? That's ultraviolet radiation working on those melanin patches.
- 12) Outline the steps of epidermal growth:
1. Along the basement membrane, cells in the stratum basale undergo mitosis. As a result, older cells are pushed upward.
 2. Cells are pushed into the stratum spinosum, where they spend most of their time creating proteins and fibers for desmosomes, which hold all the cells tightly together. They also take up melanin produced by melanocytes

in the basal layer.

3. Since nutrients must pass through the basement membrane, cells receive less and less access to them as they are pushed up. Once they reach the stratum granulosum, they are usually dead. The keratohyalin contained in the cells is the precursor to keratin.
4. By the time the cells reach the stratum corneum, they are unrecognizable. They are completely flattened and mostly just keratin.



TIP

The hair and nails are formed by this same general process so you only have to memorize it once!

- 13) Meissner's corpuscles play a role in which function? **e. The sensation of soft touch.** Although it's true that several different nerves are involved in the overall sense of touch, Meissner's corpuscles are the most responsive to touch.
- 14) Which receptor type responds to a painful stimulus? **e. Nociceptor.** Even with extreme heat or pressure, only the nociceptors can send a message that the brain interprets as painful.
- 15) What is another name for the cuticle? **c. Eponychium.** Recall that the prefix *ep-* refers to "upon" or "around," whereas the prefix *hypo-* refers to "below" or "under." The cuticle is around the base of the nail, so it's the eponychium, not the hyponychium.
- 16) The **e. sudoriferous** glands form perspiration. The Latin word *sudor* means "sweat."
- 17) The cause of graying hair is **d. lack of melanin in the shaft of the hair.** Despite the medical cause, people often suspect that answer "e. parenthood" has a lot to do with graying hair.
- 18) From where does the hair develop? **c. Follicle.** The Latin translation of this word is "small cavity" or "sac," so it makes sense that this would be an origination place.

- 19) The nails are modifications of the epidermal layers **a. corneum and lucidum**. These are the two upper layers.
- 20) Which factor is not associated with baldness? **d. Lack of melanin**. This answer just means that your hair won't turn gray, not necessarily that it will fall out of your scalp.
- 21) Sebaceous glands **b. produce an oily mixture of cholesterol, fats, and other substances**. That secretion's called *sebum* — hence “sebaceous glands.”
- 22) What function does the bulb at the base of a hair follicle serve? **c. To provide nutrients to the growing hair**. This is where connective tissue and capillaries come together to provide those nutrients.
- 23) This gland contains secretes water with fatty acids and proteins in response to stress. It acquires an unpleasant odor when bacteria breaks down the organic molecules it secretes. **a. Apocrine sweat gland**. These are the truly stinky sweat glands.



TIP Here's a memory tool for the difference between apocrine and eccrine sweat glands: You may have to APOlogize for your APOcrine glands but not your eccrine glands.

24-27) Following is how [Figure 6-2](#), the fingernail, should be labeled:

24. **b. nail bed**; 25. **a. cuticle**; 26. **d. root**; 27. **c. matrix**
- 28) When the arrector pili contracts, which gland could incidentally release its contents? **d. Sebaceous gland**. None of the other glands are the closely associated with a hair follicle; an apocrine sweat gland does secrete into the follicle but only the duct is near the follicle.

Chapter 7

A Scaffold to Build On: The Skeleton

IN THIS CHAPTER

- » Getting to know what your bones do
 - » Sorting out classifications and structures
 - » Understanding ossification
 - » Keeping the axial skeleton in line
 - » Checking out the appendicular skeleton
 - » Playing with joints
-

Human *osteology*, from the Greek word for “bone” (*osteon*) and the suffix *-logy*, which means “to study,” focuses on the 206 bones in the adult body’s endoskeleton. But osteology is more than just bones; it’s also ligaments and cartilage and the joints that make the whole assembly useful. In this chapter, you get lots of practice exploring the skeletal functions and how the joints work together.

Understanding the Functions of Dem Bones



REMEMBER The skeletal system as a whole serves five key functions:

- » **Protection:** The skeleton encases and shields delicate internal organs that might otherwise be damaged during motion or crushed by the weight of the body itself. For example, the skull’s cranium houses the brain, and the ribs and sternum of the thoracic cage protect the heart and lungs.

- » **Movement:** By providing anchor sites and a scaffold against which muscles can contract, the skeleton makes motion possible. The bones act as levers, the joints are the fulcrums, and the muscles apply the force. For instance, when the biceps muscle contracts, the radius and ulna bones of the forearm are lifted toward the humerus bone of the upper arm.
- » **Shape and Support:** The bones of the skeleton create the framework onto which the rest of our body is built, giving us our characteristic human shape. The density of the bones creates the rigidity needed to support all that tissue. The leg and foot bones are especially important for bearing the weight of our upright posture as is the curvature of our backbone.
- » **Mineral storage:** Calcium, phosphorous, and other minerals like magnesium must be maintained in the bloodstream at a constant level, so they're "banked" in the bones in case the dietary intake of those minerals drops. The bones' mineral content is constantly renewed, refreshing entirely about every nine months. A mere 35 percent decrease in blood calcium can cause convulsions.
- » **Blood cell formation:** Called *hematopoiesis*, or *hemopoiesis*, most blood cell formation in adults takes place within the red marrow inside the ends of long bones as well as within the pelvis, vertebrae, ribs, sternum, and cranial bones. Marrow produces three types of blood cells: *erythrocytes* (red cells), *leukocytes* (white cells), and *thrombocytes* (platelets). Most of these are formed in red bone marrow, although some types of white blood cells are produced in fat-rich yellow bone marrow. At birth, all bone marrow is red. With age, it converts to the yellow type. In cases of severe blood loss, the body can convert yellow marrow back to red marrow in order to increase blood cell production.

The following is an example of a question dealing with skeletal functions:



EXAMPLE Q. Which of the following is not a function of the skeleton?

- a. Support of soft tissue
- b. Hemostasis

- c. Production of red blood cells
- d. Allow for movement
- e. Mineral storage

A. The correct answer is hemostasis, which is the stoppage of bleeding or blood flow. This is one of those frequent times when study of anatomy and physiology boils down to rote memorization of Latin and Greek roots (for help, check out the Cheat Sheet at www.dummies.com/cheatsheet/anatomyphysiologywb).

1 What is happening during hematopoiesis?

- a. Bone marrow is converting from red type to yellow type.
- b. Bone marrow is forming new blood cells.
- c. Bone marrow is removing damaged bone matrix.
- d. Bone marrow is converting back from yellow type to red type.
- e. Bone marrow is releasing minerals into the blood stream.

2 Which of the following IS NOT a role of minerals like phosphorous, calcium, and magnesium?

- a. To enhance the skeleton's structural integrity
- b. To provide cushioning and nourishment to the joints
- c. To ensure availability of these minerals
- d. To form networks within the bones' structure

3 In what way does the skeleton make locomotion possible?

- a. It ensures hematopoiesis occurs as needed.
- b. Muscles use the skeleton as a scaffold against which they can contract.
- c. It cushions the joints against erythrocytes.
- d. Bones generate the motion necessary for movement within the joints.
- e. It ensures the thoracic cavity remains properly filled.

Boning Up on Classifications and Structures

Adult bones are constituted of 35 percent protein (called *ossein*), 65 percent minerals (including calcium, phosphorus, and magnesium), and a small amount of water. Minerals give the bone strength and hardness. At birth, bones are soft and pliable because they are composed mostly of cartilage. As the body grows, older cartilage is gradually replaced by hard bone tissue. Bone density, mineral content, and collagen production decreases with age, causing the bones to become more brittle and easily fractured.



REMEMBER Various types of bones make up the human skeleton, but fortunately for memorization purposes, bone type names match what the bones look like for the most part. They are as follows:

- » **Long bones**, like those found in the arms and legs, defined by their length being greater than their width.



WARNING Though they aren't very long at all, the *phalanges* (finger bones) are, in fact, long bones

- » **Short bones**, such as those in the wrists (carpals) and ankles (tarsals), have a blocky structure and allow for a greater range of motion. *Sesamoid* bones include seed-shaped bones found in joints such as the *patella*, or kneecap.
- » **Flat bones**, such as the skull, sternum, and scapulae, shield soft tissues.
- » **Irregular bones**, such as the mandible (jawbone) and vertebrae, come in a variety of shapes and sizes suited for attachment to muscles, tendons, and ligaments.

Unfortunately for students of bone structures, there's no easy way to memorize them. So brace yourself for a rapid summary of what your textbook

probably goes into in much greater detail.

Bones are comprised of two types of osseous tissue: *trabecular*, or *spongy*, bone and *compact* bone. These are found underneath the *periosteum*, which is a layer of dense irregular connective tissue that surrounds each bone. The periosteum assists in routing blood vessels deep into the bone as well as anchoring tendons and ligaments. It has both a vascular layer (the Latin word for “vessel” is *vasculum*) and an inner layer that contains the osteoblasts needed for bone growth and repair. A penetrating matrix of connective tissue, called *Sharpey’s fibers*, connects the periosteum to the bone. Long bones also have an inner membrane, the *endosteum* that creates the *medullary cavity* that houses marrow. In flat, short, and irregular bones, spongy bone is found in the middle, surrounded by a layer of compact bone then wrapped in the periosteum. Long bones are a bit more complicated as they don’t just grow in size but also change proportion.

The shaft of a long bone, the *diaphysis*, is mostly compact bone. It also contains the medullary cavity. The ends, or *epiphyses*, are encased in compact bone but contain mostly spongy bone. Between the two is the *metaphysis*, which contains the *epiphyseal plate*. This band, also referred to as the *growth plate*, is made of hyaline cartilage. This is the site of endochondral growth which allows long bones to increase in length. This process is discussed further in the next section, “[Turning Bone into Bone](#).”

Compact bone is highly organized with repeating units called *osteons*. As you read this section, refer to [Figure 7-1](#) to help you put the description with the visual. An osteon is a series of concentric circles surrounding a circular, central canal called the *Haversian canal*. Each Haversian canal contains blood and lymphatic vessels as well as nerves. Outward from there are rings of matrix called *lamellae*. Bordering each lamella are several *lacunae*, little caves housing the bone cells, or *osteocytes*. Spreading through the lamellae are tiny tunnels called *canaliculi*, which connect the lacunae. Osteocytes are able to form gap junctions (we discuss these in [Chapter 5](#)) through the canaliculi, allowing resources to be passed cell-to-cell to those far away from the Haversian canal. The Haversian canals of each osteon are connected with *Volkman’s canals*, carrying the vessels deeper into the bone.

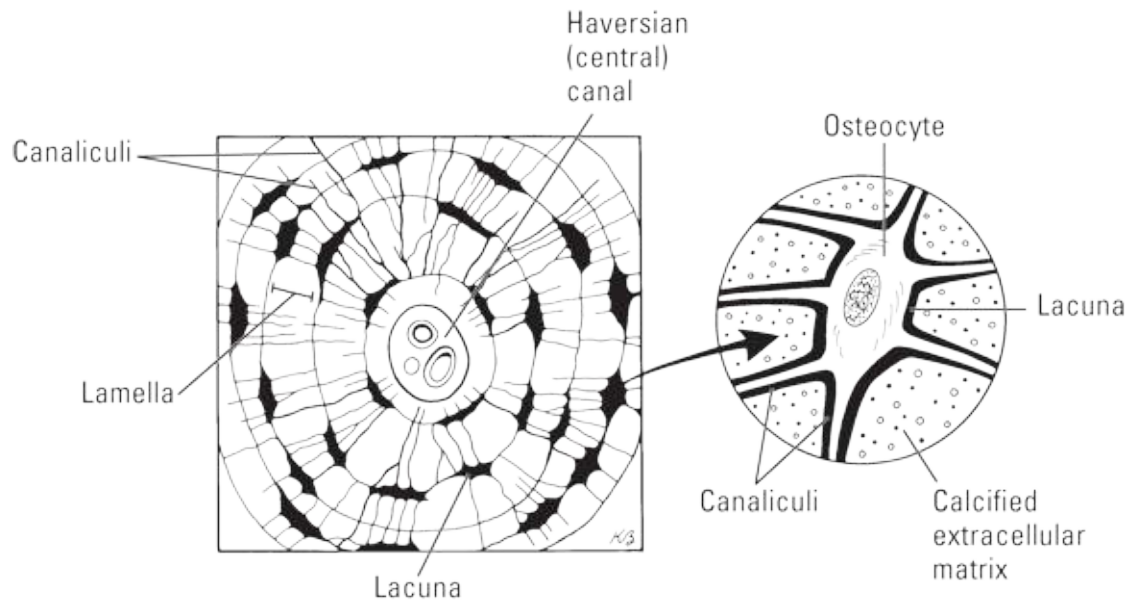


Illustration by Kathryn Born, MA

FIGURE 7-1: Compact bone tissue.

Spongy bone is much less organized. (Which means less things for you to label!) Rather than the matrix being built into rings, the calcium compounds are deposited onto collagen fibers. This forms the creation of bony bridge-like structures called *trabeculae*. While they do still have canaliculi (the little tunnels that connect cells to each other), there are no other canals or lacunae.



REMEMBER In addition to all the internal structures of a bone, there are numerous external ones. These serve a variety of purposes, such as providing a point of attachment for tendons and ligaments and creating surfaces for *articulations* (joints; *articular* is an adjective describing areas related to movement between bones). The following describes the terms used to identify these features:

- » **Process:** A broad designation for any noticeable projection (example: mastoid process)
- » **Spine:** An abrupt or pointed projection (example: scapular spine)
- » **Trochanter:** A large, usually blunt process (example: greater trochanter of femur)
- » **Tubercle:** A smaller, rounded projection (example: lesser tubercle of

humerus)

- » **Tuberosity:** A large, often rough process (example: tibial tuberosity)
- » **Crest:** A prominent ridge (example: iliac crest)
- » **Head:** A large, rounded end of a bone, often set off from the shaft by a neck (example: head of femur)
- » **Condyle:** A rounded prominence that articulates with another bone (example: medial condyle of femur)
- » **Facet:** A small, smooth, and flat surface (example: facet of thoracic vertebrae)
- » **Fossa:** A deeper depression or pit (example: mandibular fossa)
- » **Sulcus:** A shallow groove (example: scapular notch)
- » **Foramen:** A hole or opening (example: foramen magnum of occipital bone)
- » **Meatus:** A canal or tunnel (example: external acoustic meatus of temporal bone)

Let's see how well you're boning up on the skeletal structures.

- 4 Why do bones break more easily in older bodies than in younger ones?
 - a. They wear away with time and use.
 - b. Mineral content decreases, reducing their weight-bearing ability.
 - c. Older bones have larger gaps between supportive structures.
 - d. An older person's center of gravity is higher, so bones impact the ground harder in a fall.
- 5 What does the periosteum do?
 - a. It ensures circulation into even the hardest bone structures.
 - b. It allows bones to connect with ligaments and tendons.
 - c. It creates a hollowed-out space inside the bone to house marrow.
 - d. It controls calcification over time.
 - e. It allows flexibility in infant bone structures that is needed for proper

development.

6 Where are Sharpey's fibers?

- a. Inside Haversian canals
- b. Woven through the spaces in trabecular bone
- c. Between the periosteum and bone tissue
- d. Behind any lingering mastoid tissue
- e. Between cells in compact bone

7 What is inside the medullary cavity?

- a. Bone marrow
- b. Epiphyses
- c. Volkmann's canals
- d. Trabecula
- e. Osteoclasts

8 Volkmann's canals

- a. are found in compact bone only.
- b. contain the nutrient artery.
- c. pass through the epiphysis.
- d. form the center of an osteon.
- e. supply blood to articulating cartilage.

9 Which type of bones forms the weight-bearing part of the skeleton?

- a. Flat bones
- b. Irregular bones
- c. Long bones
- d. Short bones

10-18 Classify the following bones by shape. Each classification may be used more than once.

- a. Flat bone
- b. Irregular bone
- c. Long bone
- d. Short bone

- 10. _____ Vertebrae of the vertebral column
- 11. _____ Femur in thigh
- 12. _____ Sternum
- 13. _____ Tarsals in ankle
- 14. _____ Humerus in upper arm
- 15. _____ Phalanges in fingers and toes
- 16. _____ Scapulae of shoulder
- 17. _____ Kneecap
- 18. _____ Carpals in wrist

19-31 Match the description with the bone landmarks or surface features.

- a. Condyle
- b. Crest
- c. Facet
- d. Foramen
- e. Fossa
- f. Head
- g. Meatus
- h. Process
- i. Spine
- j. Sulcus
- k. Trochanter

- l. Tubercle
- m. Tuberosity

- 19. _____ An abrupt or pointed projection
- 20. _____ A large, usually blunt process
- 21. _____ A designation for any prominence or prolongation
- 22. _____ A large, often rough eminence
- 23. _____ A prominent ridge
- 24. _____ A large, rounded articular end of a bone; often set off from the shaft by the neck
- 25. _____ An oval articular prominence of a bone
- 26. _____ A smooth, flat, or nearly flat articulating surface
- 27. _____ A deeper depression
- 28. _____ A groove
- 29. _____ A hole
- 30. _____ A canal or opening to a canal
- 31. _____ A smaller, rounded eminence

32-40 Use the terms that follow to identify the regions and structures of the long bone shown in [Figure 7-2](#).

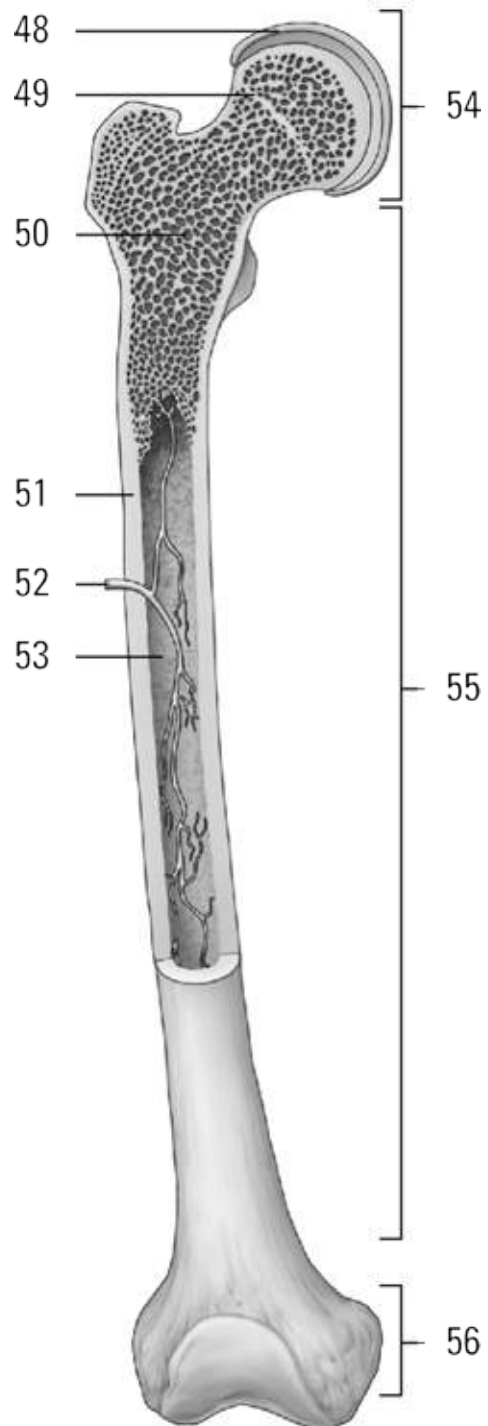


Illustration by Imagineering Media Services Inc.

FIGURE 7-2: The long bone.

- a. Diaphysis
- b. Medullary cavity
- c. Distal epiphysis